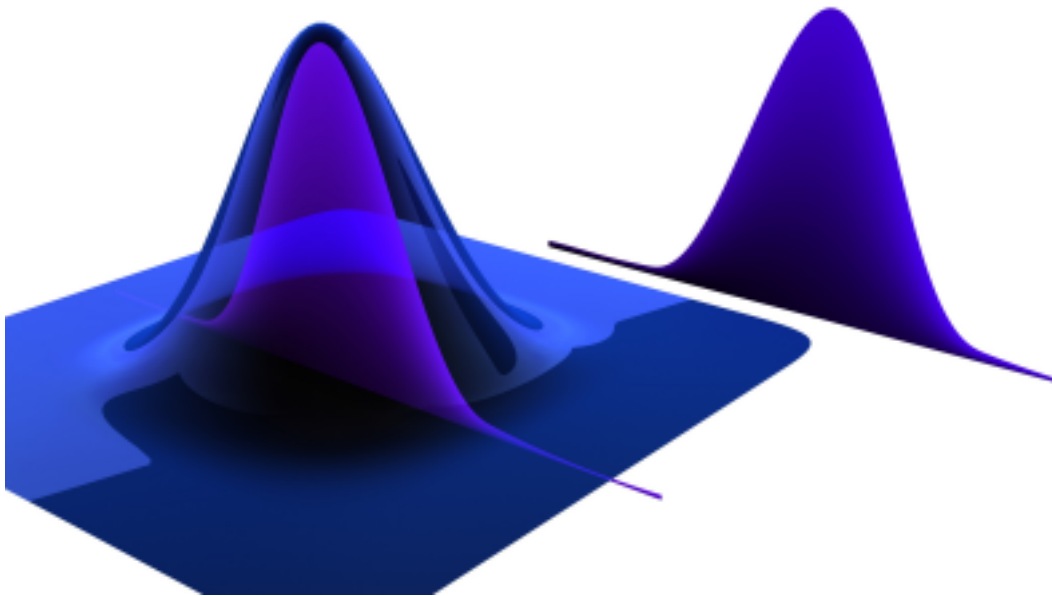


Introduction to Probability

Second Edition

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Harvard University and Stanford University
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supplementary materials available at <https://stat110.net>



To our mothers, Steffi and Min v

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Preface

This book provides a modern introduction to probability and develops a foundation for understanding statistics, randomness, and uncertainty. A variety of applications and examples are explored, from basic coin-tossing and the study of coincidences to Google PageRank and Markov chain Monte Carlo. As probability is often considered to be a counterintuitive subject, many intuitive explanations, diagrams, and practice problems are given. Each chapter ends with a section showing how to explore the ideas of that chapter in R, a free software environment for statistical calculations and simulations.

Lecture videos from Stat 110 at Harvard, the course which gave rise to this book, are freely available at <http://stat110.net>. Additional supplementary materials such as R code, animations, and solutions to the exercises marked with s, are also available at this site.

Calculus is a prerequisite for this book; there is no statistics prerequisite. The main mathematical challenge lies not in performing technical calculus derivations, but in translating between abstract concepts and concrete examples. Some major themes and features are listed below.

1. *Stories*. Throughout this book, definitions, theorems, and proofs are presented through stories: real-world interpretations that preserve mathematical precision and generality. We explore probability distributions using the generative stories that make them widely used in statistical modeling. When possible, we refrain from tedious derivations and instead aim to give interpretations and intuitions for why key results are true. Our experience is that this approach promotes long-term retention of the material by providing insight instead of demanding rote memorization.

2. *Pictures.* Since pictures are thousand-word stories, we supplement definitions with illustrations so that key concepts are associated with memorable diagrams. In many fields, the difference between a novice and an expert has been described as follows: the novice struggles to memorize a large number of seemingly disconnected facts and formulas, whereas the expert sees a unified structure in which a few principles and ideas connect these facts coherently. To help students see the structure of probability, we emphasize the connections between ideas (both verbally and visually), and at the end of most chapters we present recurring, ever-expanding maps of concepts and distributions.

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3. *Dual teaching of concepts and strategies.* Our intent is that in reading this book, students will learn not only the concepts of probability, but also a set of problem-solving strategies that are widely applicable outside of probability. In the worked examples, we explain each step of the solution but also comment on how we knew to take the approach we did. Often we present multiple solutions to the same problem.

We explicitly identify and name important strategies such as symmetry and pattern recognition, and we proactively dispel common misunderstandings, which are marked with the (biohazard) symbol.

4. *Practice problems.* The book contains about 600 exercises of varying difficulty. The exercises are intended to reinforce understanding of the material and strengthen problem-solving skills instead of requiring repetitive calculations. Some are *strategic practice problems*, grouped by theme to facilitate practice of a particular topic, while others are *mixed practice*, in which several earlier topics may need to be synthesized. About 250 exercises have detailed online solutions for practice and self-study.
5. *Simulation, Monte Carlo, and R.* Many probability problems are too difficult to solve exactly, and in any case it is important to be able to check one's answer. We introduce techniques for exploring probability via simulation, and show that often a few lines of R code suffice to create a simulation for a seemingly complicated problem.
6. *Focus on real-world relevance and statistical thinking.* Examples and exercises in this book have a clear real-world motivation, with a particular focus on building a strong foundation for further study of statistical inference and modeling. We preview important statistical ideas such as sampling, simulation, Bayesian inference, and Markov chain Monte Carlo; other application areas include genetics, medicine, computer science, and information theory. Our choice of examples and exercises is intended to highlight the power, applicability, and beauty of probabilistic thinking.

The second edition benefits from hundreds of comments, questions, and reviews from students who took courses using the book, faculty who taught with the book, and readers using the book for self-study. We have added many new examples, exercises, and explanations based on our experience teaching with the book and the feedback we have received.

New supplementary materials have also been added at <http://stat110.net>,

including animations and interactive visualizations that were created in connection with the edX online version of Stat 110. These are intended to help make probability feel more intuitive, visual, and tangible.

Preface xiii *Acknowledgments*

We thank our colleagues, the Stat 110 teaching assistants, and several thousand Stat 110 students for their comments and ideas related to the course and the book. In particular, we thank Alvin Siu, Angela Fan, Anji Tang, Anqi Zhao, Arman Sabbaghi, Carolyn Stein, David Jones, David Rosengarten, David Watson, Dennis Sun, Hyung suk Tak, Johannes Ruf, Kari Lock, Keli Liu, Kelly Bodwin, Kevin Bartz, Lazhi Wang, Martin Lysy, Michele Zemlenyi, Miles Ott, Peng Ding, Rob Phillips, Sam Fisher, Sebastian Chiu, Sofia Hou, Sushmit Roy, Theresa Gebert, Valeria Espinosa, Viktoriia Liublinska, Viviana Garcia, William Chen, and Xander Marcus.

We also thank Ella Maru Studio for helping to create the cover image for the second edition. The image illustrates the interplay between two-dimensional and one-dimensional probability distributions.

We especially thank Bo Jiang, Raj Bhuptani, Shira Mitchell, Winston Lin, and the anonymous reviewers for their detailed comments, and Andrew Gelman, Carl Morris, Persi Diaconis, Stephen Blyth, Susan Holmes, and Xiao-Li Meng for countless insightful discussions about probability.

John Kimmel at Chapman & Hall/CRC Press provided wonderful editorial expertise throughout the writing of this book. We greatly appreciate his support.

Finally, we would like to express our deepest gratitude to our families for their love and encouragement.

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Cambridge, MA and Stanford, CA
January 2019

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Probability and counting

Luck. Coincidence. Randomness. Uncertainty. Risk. Doubt. Fortune. Chance. You've probably heard these words countless times, but chances are that they were used in a vague, casual way. Unfortunately, despite its ubiquity in science and everyday life, probability can be deeply counterintuitive. If we rely on intuitions of doubtful validity, we run a serious risk of making inaccurate predictions or overconfident decisions. The goal of this book is to introduce probability as a logical framework for quantifying uncertainty and randomness in a principled way. We'll also aim to strengthen intuition, both when our initial guesses coincide with logical reasoning and when we're not so lucky.

1.1 Why study probability?

Mathematics is the logic of certainty; probability is the logic of uncertainty. Probability is extremely useful in a wide variety of fields, since it provides tools for understanding and explaining variation, separating signal from noise, and modeling complex phenomena. To give just a small sample from a continually growing list of applications:

1. *Statistics*: Probability is the foundation and language for statistics, enabling many powerful methods for using data to learn about the world.
2. *Physics*: Einstein famously said “God does not play dice with the universe”, but current understanding of quantum physics heavily involves probability at the most fundamental level of nature. Statistical mechanics is another major branch of physics that is built on probability.
3. *Biology*: Genetics is deeply intertwined with probability, both in the inheritance of genes and in modeling random mutations.
4. *Computer science*: Randomized algorithms make random choices while they are run, and in many important applications they are simpler and more efficient than any currently known deterministic alternatives. Probability also plays an essential role in studying the performance of algorithms, and in machine learning and artificial intelligence.

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5. *Meteorology*: Weather forecasts are (or should be) computed and expressed in terms of probability.
6. *Gambling*: Many of the earliest investigations of probability were aimed at answering questions about gambling and games of chance.
7. *Finance*: At the risk of redundancy with the previous example, it should be pointed out that probability is central in quantitative finance. Modeling stock prices over time and determining “fair” prices for financial instruments are based heavily on probability.
8. *Political science*: In recent years, political science has become more and more quantitative and statistical, with applications such as analyzing surveys of public opinion, assessing gerrymandering, and predicting elections.
9. *Medicine*: The development of randomized clinical trials, in which patients are randomly assigned to receive treatment or placebo, has transformed medical research in recent years. As the biostatistician David Harrington remarked, “Some have conjectured that it could be the most significant advance in scientific medicine in the twentieth century. . . . In one of the delightful ironies of modern science, the randomized trial ‘adjusts’ for both observed and unobserved heterogeneity in a controlled experiment

by introducing chance variation into the study design.” [16]

10. *Life*: Life is uncertain, and probability is the logic of uncertainty. While it isn't practical to carry out a formal probability calculation for every decision made in life, thinking hard about probability can help us avert some common fallacies, shed light on coincidences, and make better predictions.

Probability provides procedures for principled problem-solving, but it can also produce pitfalls and paradoxes. For example, we'll see in this chapter that even Gottfried Wilhelm von Leibniz and Sir Isaac Newton, the two people who independently discovered calculus in the 17th century, were not immune to basic errors in probability. Throughout this book, we will use the following strategies to help avoid potential pitfalls.

1. *Simulation*: A beautiful aspect of probability is that it is often possible to study problems via *simulation*. Rather than endlessly debating an answer with someone who disagrees with you, you can run a simulation and see empirically who is right. Each chapter in this book ends with a section that gives examples of how to do calculations and simulations in R, a free statistical computing environment.
2. *Biohazards*: Studying common mistakes is important for gaining a stronger understanding of what is and is not valid reasoning in probability. In this book, common mistakes are called *biohazards* and are denoted by $\mathcal{B}$ (since making such mistakes can be hazardous to one's health!).

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3. *Sanity checks*: After solving a problem one way, we will often try to solve the same problem in a different way or to examine whether our answer makes sense in simple and extreme cases.

1.2 Sample spaces and Pebble World

The mathematical framework for probability is built around *sets*. Imagine that an experiment is performed, resulting in one out of a set of possible outcomes. Before the experiment is performed, it is unknown which outcome will be the result; after, the result “crystallizes” into the actual outcome.

Definition 1.2.1 (Sample space and event). The *sample space* S of an experiment is the set of all possible outcomes of the experiment. An *event* A is a subset of the sample space S , and we say that A *occurred* if the actual outcome is in A .

A

B

FIGURE 1.1

A sample space as Pebble World, with two events A and B spotlighted.

The sample space of an experiment can be finite, countably infinite, or uncountably infinite (see Section A.1.5 of the math appendix for an explanation of countable and uncountable sets). When the sample space is finite, we can visualize it as *Pebble World*, as shown in Figure 1.1. Each pebble represents an outcome, and an event is a set of pebbles.

Performing the experiment amounts to randomly selecting one pebble. If all the pebbles are of the same mass, all the pebbles are equally likely to be chosen. This special case is the topic of the next two sections. In Section 1.6, we give a general definition of probability that allows the pebbles to differ in mass.

Set theory is very useful in probability, since it provides a rich language for express-

ing and working with events; Section A.1 of the math appendix provides a review of set theory. Set operations, especially unions, intersections, and complements, make it easy to build new events in terms of already-defined events. These concepts also let us express an event in more than one way; often, one expression for an event is much easier to work with than another expression for the same event.

For example, let S be the sample space of an experiment and let $A, B \subseteq S$ be events. Then the union $A \cup B$ is the event that occurs if and only if *at least one* of A, B occurs, the intersection $A \cap B$ is the event that occurs if and only if *both* A and B occur, and the complement A^c is the event that occurs if and only if A does *not* occur. We also have *De Morgan's laws*:

$$(A \cup B)^c = A^c \cap B^c \text{ and } (A \cap B)^c = A^c \cup B^c,$$

since saying that it is *not* the case that at least one of A and B occur is the same as saying that A does not occur and B does not occur, and saying that it is *not* the case that both occur is the same as saying that at least one does not occur. Analogous results hold for unions and intersections of more than two events.

In the example shown in Figure 1.1, A is a set of 5 pebbles, B is a set of 4 pebbles, $A \cup B$ consists of the 8 pebbles in A or B (including the pebble that is in both), $A \cap B$ consists of the pebble that is in both A and B , and A^c consists of the 4 pebbles that are not in A .

The notion of sample space is very general and abstract, so it is important to have some concrete examples in mind.

Example 1.2.2 (Coin flips). A coin is flipped 10 times. Writing Heads as H and Tails as T , a possible outcome (pebble) is $HHHTHHTT$, and the sample

space is the set of all possible strings of length 10 of H 's and T 's. We can (and will) encode H as 1 and T as 0, so that an outcome is a sequence $(s_1, \dots, s_{10})$ with $s_j \in \{0, 1\}$, and the sample space is the set of all such sequences. Now let's look at some events:

1. Let A_1 be the event that the first flip is Heads. As a set,

$$A_1 = \{(1, s_2, \dots, s_{10}) : s_j \in \{0, 1\} \text{ for } 2 \leq j \leq 10\}.$$

This is a subset of the sample space, so it is indeed an event; saying that A_1 occurs is the same thing as saying that the first flip is Heads. Similarly, let A_j be the event that the j th flip is Heads for $j = 2, 3, \dots, 10$.

2. Let B be the event that at least one flip was Heads. As a set,

10

$$B = \bigcup_{j=1}^{10} A_j.$$

$$A_j.$$

3. Let C be the event that all the flips were Heads. As a set,

10

$$C = \bigcap_{j=1}^{10} A_j.$$

$$A_j.$$

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consecutive Heads. As a set, 9

$$D = \bigcup_{j=1}^9 (A_j \cap A_{j+1}).$$

Example 1.2.3 (Pick a card, any card). Pick a card from a standard deck of 52 cards. The sample space S is the set of all 52 cards (so there are 52 pebbles, one for each card). Consider the following four events:

- A : card is an ace.
- B : card has a black suit.
- D : card is a diamond.
- H : card is a heart.

As a set, H consists of 13 cards:

$$\{\text{Ace of Hearts, Two of Hearts, } \dots, \text{King of Hearts}\}.$$

We can create various other events in terms of the events A , B , D , and H . Unions, intersections, and complements are especially useful for this. For example:

- $A \cap H$ is the event that the card is the Ace of Hearts.
- $A \cap B$ is the event $\{\text{Ace of Spades, Ace of Clubs}\}$.
- $A \cup D \cup H$ is the event that the card is red or an ace.
- $(A \cup B)^c = A^c \cap B^c$ is the event that the card is a red non-ace.

Also, note that $(D \cup H)^c = D^c \cap H^c = B$, so B can be expressed in terms of D and H . On the other hand, the event that the card is a spade can't be written in terms of A, B, D, H since none of them are fine-grained enough to be able to distinguish between spades and clubs.

There are *many* other events that could be defined using this sample space. In fact, the counting methods introduced later in this chapter show that there are $2^{52} \approx 4.5 \times 10^{15}$ events in this problem, even though there are only 52 pebbles.

What if the card drawn were a joker? That would indicate that we had the wrong sample space; we are assuming that the outcome of the experiment is guaranteed to be an element of S .

As the preceding examples demonstrate, events can be described in English or in set notation. Sometimes the English description is easier to interpret while the set notation is easier to manipulate. Let S be a sample space and s_{actual} be the actual outcome of the experiment (the pebble that ends up getting chosen when the experiment is performed). A mini-dictionary for converting between English and sets is given on the next page.

6

English Sets

Events and occurrences

sample space S
 s is a possible outcome $s \in S$
 A is an event $A \subseteq S$
 A occurred $s_{\text{actual}} \in A$
something must happen $s_{\text{actual}} \in S$

New events from old events

A or B (inclusive) $A \cup B$
 A and B $A \cap B$
not A A^c
 A or B , but not both $(A \cap B^c) \cup (A^c \cap B)$ at least one of $A_1, \dots$
 $A_1, A_2, \dots, A_n$ $A_1 \cup \dots \cup A_n$
all of $A_1, \dots, A_n$ $A_1 \cap \dots \cap A_n$

Relationships between events

A implies B $A \subseteq B$
 A and B are mutually exclusive $A \cap B = \emptyset$
 $A_1, \dots, A_n$ are a partition of S $A_1 \cup \dots \cup A_n = S, A_i \cap A_j = \emptyset$ for $i \neq j$

1.3 Naive definition of probability

Historically, the earliest definition of the probability of an event was to count the number of ways the event could happen and divide by the total number of possible outcomes for the experiment. We call this the *naive definition* since it is

restrictive and relies on strong assumptions; nevertheless, it is important to understand, and useful when not misused.

Definition 1.3.1 (Naive definition of probability). Let A be an event for an experiment with a finite sample space S . The *naive probability* of A is

$$P_{\text{naive}}(A) = \frac{|A|}{|S|}$$

$|S|$ = number of outcomes favorable to A
 total number of outcomes

(We use $|A|$ to denote the size of A ; see Section A.1.5 of the math appendix.)

In terms of Pebble World, the naive definition just says that the probability of A is the fraction of pebbles that are in A . For example, in Figure 1.1 it says

$P_{\text{naive}}(A) = \frac{5}{9}$, $P_{\text{naive}}(B) = \frac{4}{9}$, $P_{\text{naive}}(A \cup B) = \frac{8}{9}$, $P_{\text{naive}}(A \cap B) = \frac{1}{9}$.
Probability and counting 7 For the complements of the events just considered,

$P_{\text{naive}}(A^c) = \frac{4}{9}$, $P_{\text{naive}}(B^c) = \frac{5}{9}$, $P_{\text{naive}}((A \cup B)^c) = \frac{1}{9}$, $P_{\text{naive}}((A \cap B)^c) = \frac{8}{9}$.
 In general,

$$P_{\text{naive}}(A^c) = \frac{|S| - |A|}{|S|}$$

$$|S| - |A| = |S| - |A|$$

$$|S| = 1 - |A|$$

$$|S| = 1 - P_{\text{naive}}(A).$$

In Section 1.6, we will see that this result about complements *always* holds for probability, even when we go beyond the naive definition. A good strategy when trying to find the probability of an event is to start by thinking about whether it will be easier to find the probability of the event or the probability of its complement. De Morgan's laws are especially useful in this context, since it may be easier to work with an intersection than a union, or vice versa.

The naive definition is very restrictive in that it requires S to be finite, with equal mass for each pebble. It has often been misapplied by people who assume equally likely outcomes without justification and make arguments to the effect of "either it will happen or it won't, and we don't know which, so it's 50-50". In addition to sometimes giving absurd probabilities, this type of reasoning isn't even internally consistent. For example, it would say that the probability of life on Mars is $1/2$ ("either there is or there isn't life there"), but it would also say that the probability of *intelligent* life on Mars is $1/2$, and it is clear intuitively—and by the properties of probability developed in Section 1.6—that the latter should have strictly lower probability than the former. But there are several important types of problems where the naive definition is applicable:

- when there is *symmetry* in the problem that makes outcomes equally likely. It is common to assume that a coin has a 50% chance of landing Heads when tossed, due to the physical symmetry of the coin.¹ For a standard, well-shuffled deck of cards, it is reasonable to assume that all orders are equally likely. There aren't certain overeager cards that especially like to be near the top of the deck; any particular location in the deck is equally likely to house any of the 52 cards.

- when the outcomes are equally likely *by design*. For example, consider conducting a survey of n people in a population of N people. A common goal is to obtain a *simple random sample*, which means that the n people are chosen randomly with all subsets of size n being equally likely. If successful, this ensures that the naive definition is applicable, but in practice this may be hard to accomplish because of various complications, such as not having a complete, accurate list of contact information for everyone in the population.

¹See Diaconis, Holmes, and Montgomery [7] for a physical argument that the chance of a tossed coin coming up the way it started is about 0.51 (close to but slightly more than 1/2), and Gelman and Nolan [11] for an explanation of why the probability of Heads is close to 1/2 even for a coin that is manufactured to have different weights on the two sides (for standard coin-tossing; allowing the coin to spin is a different matter).

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- when the naive definition serves as a useful *null model*. In this setting, we *assume* that the naive definition applies just to see what predictions it would yield, and then we can compare observed data with predicted values to assess whether the hypothesis of equally likely outcomes is tenable.

1.4 How to count

Calculating the naive probability of an event A involves counting the number of pebbles in A and the number of pebbles in the sample space S . Often the sets we need to count are extremely large. This section introduces some fundamental methods for counting; further methods can be found in books on *combinatorics*, the branch of mathematics that studies counting.

1.4.1 Multiplication rule

In some problems, we can directly count the number of possibilities using a basic but versatile principle called the *multiplication rule*. We'll see that the multiplication rule leads naturally to counting rules for *sampling with replacement* and *sampling without replacement*, two scenarios that often arise in probability and statistics.

Theorem 1.4.1 (Multiplication rule). Consider a compound experiment consisting of two sub-experiments, Experiment A and Experiment B. Suppose that Experiment A has a possible outcomes, and for each of those outcomes Experiment B has b possible outcomes. Then the compound experiment has ab possible outcomes.

To see why the multiplication rule is true, imagine a tree diagram as in Figure 1.2. Let the tree branch a ways according to the possibilities for Experiment A, and for each of those branches create b further branches for Experiment B. Overall, there are $b + b + \cdots + b$ possibilities. Since there are a such sums, the total is ab possibilities.

1.4.2. It is often easier to think about the experiments as being in chronological

order, but there is no requirement in the multiplication rule that Experiment A has to be performed before Experiment B.

Example 1.4.3 (Runners). Suppose that 10 people are running a race. Assume that ties are not possible and that all 10 will complete the race, so there will be well defined first place, second place, and third place winners. How many possibilities are there for the first, second, and third place winners?

Solution: There are 10 possibilities for who gets first place, then once that is fixed there are 9 possibilities for who gets second place, and once these are both fixed

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FIGURE 1.2

Tree diagram illustrating the multiplication rule. If Experiment A has 3 possible outcomes, for each of which Experiment B has 4 possible outcomes, then overall there are $3 \cdot 4 = 12$ possible outcomes.

there are 8 possibilities for third place. So by the multiplication rule, there are $10 \cdot 9 \cdot 8 = 720$ possibilities.

We did not have to consider the first place winner first. We could just as well have said that there are 10 possibilities for who got third place, then once that is fixed there are 9 possibilities for second place, and once those are both fixed there are 8 possibilities for first place. Or imagine that there are 3 platforms, which the first, second, and third place runners will stand on after the race. The platforms are gold, silver, and bronze, allocated to the first, second, and third place runners, respectively. Again there are $10 \cdot 9 \cdot 8 = 720$ possibilities for how the platforms will be occupied after the race, and there is no reason that the platforms must be considered in the order (gold, silver, bronze).

Example 1.4.4 (Chessboard). How many squares are there in an 8×8 chessboard, as in Figure 1.3? Even the name " 8×8 chessboard" makes this easy: there are $8 \cdot 8 = 64$ squares on the board. The grid structure makes this clear, but we can also think of this as an example of the multiplication rule: to specify a square, we can specify which row and which column it is in. There are 8 choices of row, for each of which there are 8 choices of column.

Furthermore, we can see without doing any calculations that half the squares are

white and half are black. Imagine rotating the chessboard 90 degrees clockwise. Then all the positions that had a white square now contain a black square, and vice versa, so the number of white squares must equal the number of black squares. We

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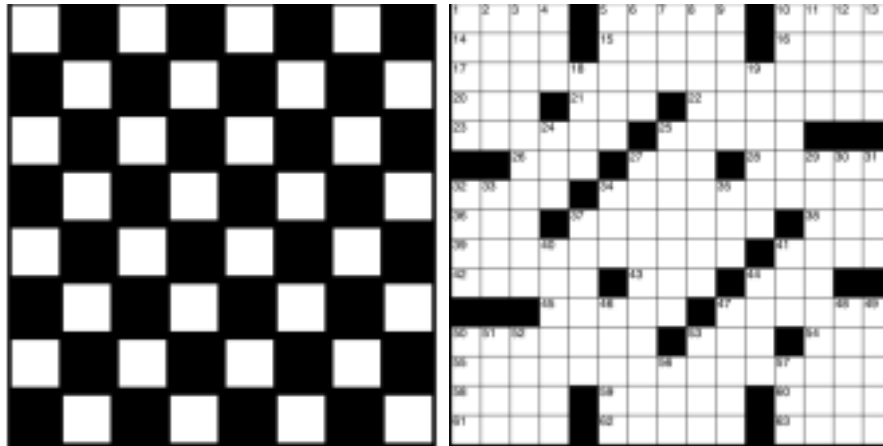


FIGURE 1.3

An 8×8 chessboard (left) and a crossword puzzle grid (right). The chessboard has $8 \cdot 8 = 64$ squares, whereas counting the number of white squares in the crossword puzzle grid requires more work.

can also count the number of white squares using the multiplication rule: in each of the 8 rows there are 4 white squares, giving a total of $8 \cdot 4 = 32$ white squares.

In contrast, it would require more effort to count the number of white squares in the crossword puzzle grid shown in Figure 1.3. The multiplication rule does not apply, since different rows sometimes have different numbers of white squares.

Example 1.4.5 (Ice cream cones). Suppose you are buying an ice cream cone. You can choose whether to have a cake cone or a waffle cone, and whether to have chocolate, vanilla, or strawberry as your flavor. This decision process can be visualized with a tree diagram, as in Figure 1.4.

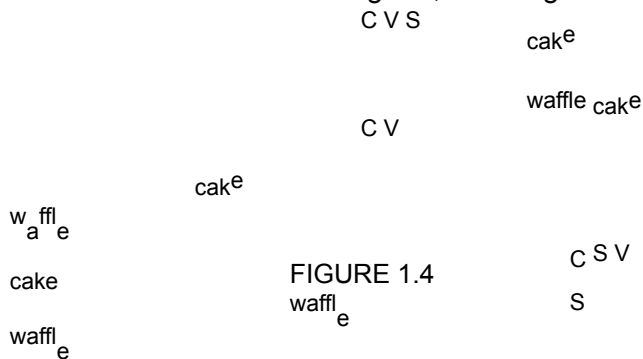


FIGURE 1.4

Tree diagram for choosing an ice cream cone. Regardless of whether the type of cone or the flavor is chosen first, there are $2 \cdot 3 = 3 \cdot 2 = 6$ possibilities.

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By the multiplication rule, there are $2 \cdot 3 = 6$ possibilities. This is a very simple example, but is worth thinking through in detail as a foundation for thinking about

and visualizing more complicated examples. Soon we will encounter examples where drawing the tree in a legible size would take up more space than exists in the known universe, yet where conceptually we can still think in terms of the ice cream example. Some things to note:

1. It doesn't matter whether you choose the type of cone first ("I'd like a waffle cone with chocolate ice cream") or the flavor first ("I'd like chocolate ice cream on a waffle cone"). Either way, there are $2 \cdot 3 = 3 \cdot 2 = 6$ possibilities.
2. It doesn't matter whether the same flavors are available on a cake cone as on a waffle cone. What matters is that there are exactly 3 flavor choices for each cone choice. If for some strange reason it were forbidden to have chocolate ice cream on a waffle cone, with no substitute flavor available (aside from vanilla and strawberry), there would be $3 + 2 = 5$ possibilities and the multiplication rule wouldn't apply. In larger examples, such complications could make counting the number of possibilities vastly more difficult.

Now suppose you buy *two* ice cream cones on a certain day, one in the afternoon and the other in the evening. Write, for example, (cakeC, waffleV) to mean a cake cone with chocolate in the afternoon, followed by a waffle cone with vanilla in the evening. By the multiplication rule, there are $6^2 = 36$ possibilities in your delicious compound experiment.

But what if you're only interested in what kinds of ice cream cones you had that day, not the order in which you had them, so you don't want to distinguish, for example, between (cakeC, waffleV) and (waffleV, cakeC)? Are there now $36/2 = 18$ possibilities? No, since possibilities like (cakeC, cakeC) were already only listed once each. There are $6 \cdot 5 = 30$ ordered possibilities (x, y) with $x \neq y$, which turn into 15 possibilities if we treat (x, y) as equivalent to (y, x) , plus 6 possibilities of the form (x, x) , giving a total of 21 possibilities. Note that if the 36 original ordered pairs (x, y) are equally likely, then the 21 possibilities here are *not* equally likely.

Example 1.4.6 (Subsets). A set with n elements has 2^n subsets, including the empty set $\emptyset$ and the set itself. This follows from the multiplication rule since for each element, we can choose whether to include it or exclude it. For example, the set $\{1, 2, 3\}$ has the 8 subsets $\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}$. This result explains why in Example 1.2.3 there are 2^{52} events that can be defined.

We can use the multiplication rule to arrive at formulas for sampling with and with out replacement. Many experiments in probability and statistics can be interpreted in one of these two contexts, so it is appealing that both formulas follow directly from the same basic counting principle.

Theorem 1.4.7 (Sampling with replacement). Consider n objects and making k choices from them, one at a time *with replacement* (i.e., choosing a certain object does not preclude it from being chosen again). Then there are n^k possible outcomes

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(where order matters, in the sense that, e.g., choosing object 3 and then object 7 is counted as a different outcome than choosing object 7 and then object 3.)

For example, imagine a jar with n balls, labeled from 1 to n . We sample balls one at a time with replacement, meaning that each time a ball is chosen, it is returned to the jar. Each sampled ball is a sub-experiment with n possible outcomes, and

there are k sub-experiments. Thus, by the multiplication rule there are n^k ways to obtain a sample of size k .

Theorem 1.4.8 (Sampling without replacement). Consider n objects and making k choices from them, one at a time *without replacement* (i.e., choosing a certain object precludes it from being chosen again). Then there are $n(n-1) \cdot \dots \cdot (n-k+1)$ possible outcomes for $1 \leq k \leq n$, and 0 possibilities for $k > n$ (where order matters). By convention, $n(n-1) \cdot \dots \cdot (n-k+1) = n$ for $k = 1$.

This result also follows directly from the multiplication rule: each sampled ball is again a sub-experiment, and the number of possible outcomes decreases by 1 each time. Note that for sampling k out of n objects without replacement, we need $k \leq n$, whereas in sampling with replacement the objects are inexhaustible.

Example 1.4.9 (Permutations and factorials). A *permutation* of $1, 2, \dots, n$ is an arrangement of them in some order, e.g., 3, 5, 1, 2, 4 is a permutation of 1, 2, 3, 4, 5. By Theorem 1.4.8 with $k = n$, there are $n!$ permutations of $1, 2, \dots, n$. For example, there are $n!$ ways in which n people can line up for ice cream. (Recall that $n!$ is $n(n-1)(n-2) \cdot \dots \cdot 1$ for any positive integer n , and $0! = 1$.)

Theorems 1.4.7 and 1.4.8 are theorems about *counting*, but when the naive definition applies, we can use them to calculate *probabilities*. This brings us to our next example, a famous problem in probability called the *birthday problem*. The solution incorporates both sampling with replacement and sampling without replacement.

Example 1.4.10 (Birthday problem). There are k people in a room. Assume each person's birthday is equally likely to be any of the 365 days of the year (we exclude February 29), and that people's birthdays are independent (we will define *independence* formally later, but intuitively it means that knowing some people's birthdays gives us no information about other people's birthdays; this would not hold if, e.g., we knew that two of the people were twins). What is the probability that at least one pair of people in the group have the same birthday?

Solution:

There are 365^k ways to assign birthdays to the people in the room, since we can imagine the 365 days of the year being sampled k times, with replacement. By assumption, all of these possibilities are equally likely, so the naive definition of probability applies.

Used directly, the naive definition says we just need to count the number of ways to assign birthdays to k people such that there are two people who share a birthday.

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But this counting problem is hard, since it could be Emma and Steve who share a birthday, or Steve and Naomi, or all three of them, or the three of them could share a birthday while two others in the group share a different birthday, or various other possibilities.

Instead, let's count the complement: the number of ways to assign birthdays to k people such that no two people share a birthday. This amounts to sampling the 365 days of the year *without* replacement, so the number of possibilities is $365 \cdot 364 \cdot 363 \cdot \dots \cdot (365 - k + 1)$ for $k \leq 365$. Therefore the probability of no birthday matches in a group of k people is

$$P(\text{no birthday match}) = \frac{365 \cdot 364 \cdot \dots \cdot (365 - k + 1)}{365^k},$$

and the probability of at least one birthday match is

$$P(\text{at least 1 birthday match}) = 1 - \frac{365 \cdot 364 \cdot \dots \cdot (365 - k + 1)}{365^k}.$$

Figure 1.5 plots the probability of at least one birthday match as a function of k . The first value of k for which the probability of a match exceeds 0.5 is $k = 23$. Thus, in a group of 23 people, there is a better than 50% chance that there is at least one birthday match. At $k = 57$, the probability of a match already exceeds 99%.

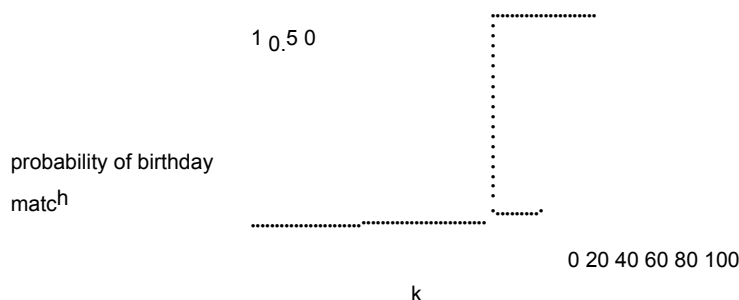


FIGURE 1.5

Probability that in a room of k people, at least two were born on the same day. This probability first exceeds 0.5 when $k = 23$.

Of course, for $k = 366$ we are *guaranteed* to have a match, but it's surprising that even with a much smaller number of people it's overwhelmingly likely that there is a birthday match. For a quick intuition into why it should not be so surprising, note that with 23 people there are $\binom{23}{2} = 253$ *pairs* of people, any of which could be a birthday match.

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Problems 26 and 27 show that the birthday problem is much more than a fun party game, and much more than a way to build intuition about coincidences; there are also important applications in statistics and computer science. Problem 62 explores the more general setting in which the probability is not necessarily $1/365$ for each day. It turns out that in the non-equal probability case, having at least one match becomes even *more* likely.

1.4.11 (Labeling objects). Drawing a sample from a population is a very fundamental concept in statistics. It is important to think of the objects or people in the population as *named* or *labeled*. For example, if there are n balls in a jar, we can imagine that they have labels from 1 to n , even if the balls look the same to the human eye. In the birthday problem, we can give each person an ID (identification) number, rather than thinking of the people as indistinguishable particles or a faceless mob.

A related example is an instructive blunder made by Leibniz in a seemingly simple problem (see Gorroochurn [14] for discussion of this and a variety of other probability problems from a historical perspective).

Example 1.4.12 (Leibniz's mistake). If we roll two fair dice, which is more likely: a sum of 11 or a sum of 12?

Solution:

Label the dice A and B, and consider each die to be a sub-experiment. By the multiplication rule, there are 36 possible outcomes for ordered pairs of the form (value of A, value of B), and they are equally likely by symmetry. Of these, (5, 6) and (6, 5) are favorable to a sum of 11, while only (6, 6) is favorable to a sum of 12. Therefore a sum of 11 is twice as likely as a sum of 12; the probability is $1/18$ for the former, and $1/36$ for the latter.

However, Leibniz wrongly argued that a sum of 11 and a sum of 12 are equally likely, claiming that each of these sums can be attained in only one way. Here Leibniz was making the mistake of treating the two dice as indistinguishable objects, viewing (5, 6) and (6, 5) as the same outcome.

What are the antidotes to Leibniz's mistake? First, as explained in 1.4.11, we should *label* the objects in question instead of treating them as indistinguishable. If Leibniz had labeled his dice A and B, or green and orange, or left and right, he would not have made this mistake. Second, before we use counting for probability, we should ask ourselves whether the naive definition applies (see 1.4.23 for another example showing that caution is needed before applying the naive definition).

1.4.2 Adjusting for overcounting

In many counting problems, it is not easy to directly count each possibility once and only once. If, however, we are able to count each possibility exactly c times

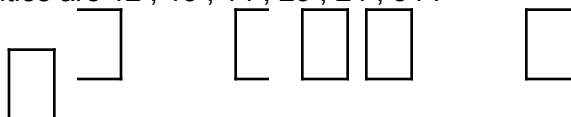
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for some c , then we can adjust by dividing by c . For example, if we have exactly double-counted each possibility, we can divide by 2 to get the correct count. We call this *adjusting for overcounting*.

Example 1.4.13 (Committees and teams). Consider a group of four people.

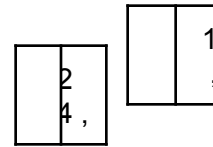
(a) How many ways are there to choose a two-person committee? (b) How many ways are there to break the people into two teams of two? *Solution:*

(a) One way to count the possibilities is by listing them out: labeling the people as 1, 2, 3, 4, the possibilities are 12, 13, 14, 23, 24, 34.



Another approach is to use the multiplication rule with an adjustment for overcounting. By the multiplication rule, there are 4 ways to choose the first person on the committee and 3 ways to choose the second person on the committee, but this counts each possibility twice, since picking 1 and 2 to be on the committee is the same as picking 2 and 1 to be on the committee. Since we have overcounted by a factor of 2, the number of possibilities is $(4 \cdot 3)/2 = 6$.

(b) Here are 3 ways to see that there are 3 ways to form the teams. Labeling the people as 1, 2, 3, 4, we can directly list out the possibilities:



Listing out all possibilities would quickly become tedious or infeasible with



more people though. Another approach is to note that it suffices to specify person 1's teammate (and then the other team is determined). A third way is to use (a) to see that there are 6 ways to choose one team. This overcounts by a factor of 2, since picking 1 and 2 to be a team is equivalent to picking 3 and 4 to be a team. So again the answer is $6/2 = 3$.

A *binomial coefficient* counts the number of subsets of a certain size for a set, such as the number of ways to choose a committee of size k from a set of n people. Sets and subsets are by definition *unordered*, e.g., $\{3, 1, 4\} = \{4, 1, 3\}$, so we are counting the number of ways to choose k objects out of n , without replacement and without distinguishing between the different orders in which they could be chosen.

Definition 1.4.14 (Binomial coefficient). For any nonnegative integers k and n , the *binomial coefficient* $\binom{n}{k}$, read as “ n choose k ”, is the number of subsets of size k for a set of size n .

For example, $\binom{4}{2} = 6$, as shown in Example 1.4.13. The binomial coefficient $\binom{n}{k}$ is sometimes called a *combination*, but we do not use that terminology here since “combination” is such a useful general-purpose word. Algebraically, binomial coefficients can be computed as follows.

Theorem 1.4.15 (Binomial coefficient formula). For $k \leq n$, we have

$$\binom{n}{k} = \frac{n(n-1) \cdots (n-k+1)}{k!} = \frac{n!}{k!(n-k)!}$$

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For $k > n$, we have $\binom{n}{k} = 0$.

Proof. Let A be a set with $|A| = n$. Any subset of A has size at most n , so $\binom{n}{k} = 0$ for $k > n$. Now let $k \leq n$. By Theorem 1.4.8, there are $n(n-1) \cdots (n-k+1)$ ways to make an *ordered* choice of k elements without replacement. This overcounts each subset of interest by a factor of $k!$ (since we don't care how these elements are ordered), so we can get the correct count by dividing by $k!$.

1.4.16. The binomial coefficient $\binom{n}{k}$ is often defined in terms of factorials, but keep in mind that $\binom{n}{k}$ is 0 if $k > n$, even though the factorial of a negative number is

undefined. Also, the middle expression in Theorem 1.4.15 is often better for computation than the expression with factorials, since factorials grow *extremely* fast. For example,

$$100 \quad = 100 \cdot 99$$

$$2 \quad = 4950$$

can even be done by hand, whereas computing $\binom{100}{2} = 100!/(98! \cdot 2!)$ by first calculating $100!$ and $98!$ would be wasteful and possibly dangerous because of the extremely large numbers involved ($100! \approx 9.33 \times 10^{157}$).

Example 1.4.17 (Club officers). In a club with n people, there are $n(n-1)(n-2)$ ways to choose a president, vice president, and treasurer, and there are

$$\frac{n(n-1)(n-2)}{3!}$$

$$3$$

ways to choose 3 officers without predetermined titles.

Example 1.4.18 (Permutations of a word). How many ways are there to permute the letters in the word LALALAAA? To determine a permutation, we just need to choose where the 5 A's go (or, equivalently, just decide where the 3 L's go). So there

$$\frac{8!}{5!} = \frac{8 \cdot 7 \cdot 6}{3!} = 56 \text{ permutations.}$$

How many ways are there to permute the letters in the word STATISTICS? Here are two approaches. We could choose where to put the S's, then where to put the T's (from the remaining positions), then where to put the I's, then where to put the A (and then the C is determined). Alternatively, we can start with $10!$ and then adjust for overcounting, dividing by $3!3!2!$ to account for the fact that the S's can be permuted among themselves in any way, and likewise for the T's and I's. This gives

$$\frac{10!}{3!3!2!} = \frac{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{2 \cdot 1 \cdot 1} = 50400 \text{ possibilities.}$$

Example 1.4.19 (Binomial theorem). The *binomial theorem* states that

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k},$$

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for any nonnegative integer n . To prove the binomial theorem, expand out the product

$$(x + y)(x + y) \cdots (x + y)$$

n factors

Just as $(a + b)(c + d) = ac + ad + bc + bd$ is the sum of terms where we pick the a or the b from the first factor (but not both) and the c or the d from the second factor (but not both), the terms of $(x + y)^n$ are obtained by picking either the x or the y (but not both) from each factor. There are $\binom{n}{k}$ ways to choose exactly k of the x 's, and each such choice yields the term $x^k y^{n-k}$. The binomial theorem follows.

We can use binomial coefficients to calculate probabilities in many problems for which the naive definition applies.

Example 1.4.20 (Full house in poker). A 5-card hand is dealt from a standard, well-shuffled 52-card deck. The hand is called a *full house* in poker if it consists of three cards of some rank and two cards of another rank, e.g., three 7's and two 10's (in any order). What is the probability of a full house?

Solution:

All of the $\binom{52}{5}$ possible hands are equally likely by symmetry, so the naive definition is applicable. To find the number of full house hands, use the multiplication rule (and imagine the tree). There are 13 choices for what rank we have three of; for concreteness, assume we have three 7's and focus on that branch of the tree. There are $\binom{4}{3}$ ways to choose which 7's we have. Then there are 12 choices for what rank we have two of, say 10's for concreteness, and $\binom{4}{2}$ ways to choose two 10's. Thus,

$$P(\text{full house}) = \frac{13 \binom{4}{3} 12 \binom{4}{2}}{\binom{52}{5}} = \frac{3744}{2598960} \approx 0.00144.$$

The decimal approximation is more useful when playing poker, but the answer in terms of binomial coefficients is exact and *self-annotating* (seeing $\binom{52}{5}$ is a much bigger hint about its origin than seeing 2598960).

Example 1.4.21 (Newton-Pepys problem). Isaac Newton was consulted about the following problem by Samuel Pepys, who wanted the information for gambling purposes. Which of the following events has the highest probability?

A: At least one 6 appears when 6 fair dice are rolled.

B: At least two 6's appear when 12 fair dice are rolled.

C: At least three 6's appear when 18 fair dice are rolled.

Solution:

The three experiments have 6^6 , 6^{12} , and 6^{18} possible outcomes, respectively, and by symmetry the naive definition applies in all three experiments.

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A: Instead of counting the number of ways to obtain at least one 6, it is easier to

count the number of ways to get no 6's. Getting no 6's is equivalent to sampling the numbers 1 through 5 with replacement 6 times, so 5^6 outcomes are favorable to A^c (and $6^6 - 5^6$ are favorable to A). Thus

$$P(A) = 1 - \frac{5^6}{6^6} \approx 0.67.$$

B : Again we count the outcomes in B^c first. There are 5^{12} ways to get no 6's in 12 die rolls. There are $^{12}_1 5^{11}$ ways to get exactly one 6: we first choose which die lands 6, then sample the numbers 1 through 5 with replacement for the other 11 dice. Adding these, we get the number of ways to fail to obtain at least two 6's. Then

$$P(B) = 1 - \frac{5^{12} + ^{12}_1 5^{11}}{6^{12}} \approx$$

C : We count the outcomes in C^c , i.e., the number of ways to get zero, one, or two 6's in 18 die rolls. There are 5^{18} ways to get no 6's, $^{18}_1 5^{17}$ ways to get exactly one 6, and $^{18}_2 5^{16}$ ways to get exactly two 6's (choose which two dice will land 6, then decide how the other 16 dice will land).

$$P(C) = 1 - \frac{5^{18} + ^{18}_1 5^{17} + ^{18}_2 5^{16}}{6^{18}} \approx 0.60.$$

Therefore A has the highest probability.

Newton arrived at the correct answer using similar calculations. Newton also provided Pepys with an intuitive argument for why A was the most likely of the three; however, his intuition was invalid. As explained in Stigler [24], using loaded dice could result in a different ordering of A , B , C , but Newton's intuitive argument did not depend on the dice being fair.

In this book, we care about counting not for its own sake, but because it sometimes helps us to find probabilities. Here is an example of a neat but treacherous counting problem; the solution is elegant, but it is rare that the result can be used with the naive definition of probability.

Example 1.4.22 (Bose-Einstein). How many ways are there to choose k times from a set of n objects with replacement, if order doesn't matter (we only care about how many times each object was chosen, not the order in which they were chosen)?

Solution:

When order does matter, the answer is n^k by the multiplication rule, but this problem is much harder. We will solve it by solving an *isomorphic* problem (the same problem in a different guise).

Let us find the number of ways to put k indistinguishable particles into n distinguishable boxes. That is, swapping the particles in any way is not considered a separate

possibility: all that matters are the *counts* for how many particles are in each box.

This scenario is known as a *Bose-Einstein* problem, since the physicists Satyendra Nath Bose and Albert Einstein studied related problems about indistinguishable particles in the 1920s, using their ideas to successfully predict the existence of a strange state of matter known as a Bose-Einstein condensate.

Any configuration can be encoded as a sequence of |'s and •'s in a natural way, as illustrated in Figure 1.6.

FIGURE 1.6

Bose-Einstein encoding: putting $k = 7$ indistinguishable particles into $n = 4$ distinguishable boxes can be expressed as a sequence of |'s and •'s, where | denotes a wall and • denotes a particle.

To be valid, a sequence must start and end with a |, and have exactly $n - 1$ |'s and exactly k •'s in between the starting and ending |'s; conversely, any such sequence is a valid encoding for some configuration of particles in boxes. Imagine that we have written down the starting and ending |'s, which represent the outer walls, and in between there are $n + k - 1$ fill-in-the-blank slots in between the outer walls. We need only choose where to put the k •'s (since then where the $n + k - 1$ interior |'s go is completely determined). So the number of possibilities is $\binom{n+k-1}{k}$. This

counting method is sometimes called the *stars and bars* argument, where here we have dots in place of stars.

To relate this result back to the original question, we can let each box correspond to one of the n objects and use the particles as “check marks” to tally how many times each object is selected. For example, if a certain box contains exactly 3 particles, that means the object corresponding to that box was chosen exactly 3 times. The particles being indistinguishable corresponds to the fact that we don't care about the order in which the objects are chosen. Thus, the answer to the original question is also $\binom{n+k-1}{k}$.

k

Another isomorphic problem is to count the number of solutions $(x_1, \dots, x_n)$ to the equation $x_1 + x_2 + \dots + x_n = k$, where the x_i are nonnegative integers. This is equivalent since we can think of x_i as the number of particles in the i th box.

1.4.23. The Bose-Einstein result should *not* be used in the naive definition of probability except in very special circumstances. For example, consider a survey where a sample of size k is collected by choosing people from a population of size n

one at a time, with replacement and with equal probabilities. Then the n^k *ordered* samples are equally likely, making the naive definition applicable, but the

unordered samples (where all that matters is how many times each person was sampled) are *not* equally likely.

As another example, with $n = 365$ days in a year and k people, how many possible *unordered* birthday lists are there? For example, for $k = 3$, we want to count lists like (May 1, March 31, April 11), where all permutations are considered equivalent. We can't do a simple adjustment for overcounting such as $n^k/3!$ since, e.g., there are 6 permutations of (May 1, March 31, April 11) but only 3

permutations of (March

31, March 31, April 11). By

Bose-Einstein, the number of lists is

$\binom{n+k-1}{k}$

k

. But the

ordered birthday lists are equally likely, not the unordered lists, so the

Bose-Einstein value should not be used in calculating birthday probabilities.

1.5 Story proofs

A *story proof* is a proof by interpretation. For counting problems, this often means counting the same thing in two different ways, rather than doing tedious algebra. A story proof often avoids messy calculations and goes further than an algebraic proof toward *explaining* why the result is true. The word “story” has several meanings, some more mathematical than others, but a story proof (in the sense in which we’re using the term) is a fully valid mathematical proof. Here are some examples of story proofs, which also serve as further examples of counting.

Example 1.5.1 (Choosing the complement). For any nonnegative integers n and k with $k \leq n$, we have

$$\binom{n}{k} = \binom{n}{n-k}.$$

This is easy to check algebraically (by writing the binomial coefficients in terms of factorials), but a story proof makes the result easier to understand intuitively.

Story proof : Consider choosing a committee of size k in a group of n people. We know that there are $\binom{n}{k}$ possibilities. But another way to choose the committee is to specify which $n - k$ people are *not* on the committee; specifying who is on the committee determines who is *not* on the committee, and vice versa. So the two sides are equal, as they are two ways of counting the same thing.

Example 1.5.2 (The team captain). For any positive integers n and k with $k \leq n$,

$$\binom{n-1}{k-1} = \frac{n}{k} \binom{n-1}{k}.$$

This is again easy to check algebraically (using the fact that $m! = m(m-1)!$) for

any positive integer m), but a story proof is more insightful.

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Story proof : Consider a group of n people, from which a team of k will be chosen, one of whom will be the team captain. To specify a possibility, we could first choose the team captain and then choose the remaining $k - 1$ team members; this gives the left-hand side. Equivalently, we could first choose the k team members and then choose one of them to be captain; this gives the right-hand side.

Example 1.5.3 (Vandermonde's identity). A famous relationship between binomial coefficients, called *Vandermonde's identity*,² says that

$$\binom{m+n}{k} = \sum_{j=0}^k \binom{m}{j} \binom{n}{k-j}.$$

This identity will come up several times in this book. Trying to prove it with a brute force expansion of all the binomial coefficients would be a nightmare. But a story proves the result elegantly and makes it clear *why* the identity holds.

Story proof : Consider a student organization consisting of m juniors and n seniors, from which a committee of size k will be chosen. There are $\binom{m+n}{k}$ possibilities.

If there are j juniors in the committee, then there must be $k - j$ seniors in the committee. The right-hand side of the identity sums up the cases for j .

Example 1.5.4 (Partnerships). Let's use a story proof to show that

$$(2n)! = 2^n \cdot n! = (2n-1)(2n-3) \cdots 3 \cdot 1.$$

Story proof : We will show that both sides count the number of ways to break $2n$ people into n partnerships. Take $2n$ people, and give them ID numbers from 1 to $2n$. We can form partnerships by lining up the people in some order and then saying the first two are a pair, the next two are a pair, etc. This overcounts by a factor of $n! \cdot 2^n$ since the order of pairs doesn't matter, nor does the order within each pair. Alternatively, count the number of possibilities by noting that there are $2n-1$ choices for the partner of person 1, then $2n-3$ choices for person 2 (or person 3, if person 2 was already paired to person 1), and so on.

1.6 Non-naive definition of probability

We have now seen several methods for counting outcomes in a sample space, allowing us to calculate probabilities if the naive definition applies. But the naive definition can only take us so far, since it requires equally likely outcomes and can't handle

²Vandermonde's identity is named after the 18th century French mathematician Alexandre

Théophile Vandermonde, but it was discovered much earlier, and stated in 1303 by the Chinese mathematician Zhu Shijie.

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an infinite sample space. To generalize the notion of probability, we'll write down a short wish list of how we want probability to behave (in math, the items on the wish list are called *axioms*), and then define a probability function to be something that satisfies the properties we want!

Here is the general definition of probability that we'll use for the rest of this book. It requires just two axioms, but from these axioms it is possible to prove a vast array of results about probability.

Definition 1.6.1 (General definition of probability). A *probability space* consists of a sample space S and a *probability function* P which takes an event $A \subseteq S$ as input and returns $P(A)$, a real number between 0 and 1, as output. The function P must satisfy the following axioms:

1. $P(\emptyset) = 0$, $P(S) = 1$.

2. If $A_1, A_2, \dots$ are disjoint events, then

$$P\left(\bigcup_{j=1}^{\infty} A_j\right) = \sum_{j=1}^{\infty} P(A_j).$$

(Saying that these events are *disjoint* means that they are *mutually exclusive*: $A_i \cap A_j = \emptyset$ for $i \neq j$.)

In Pebble World, the definition says that probability behaves like mass: the mass of an empty pile of pebbles is 0, the total mass of all the pebbles is 1, and if we have non-overlapping piles of pebbles, we can get their combined mass by adding the masses of the individual piles. Unlike in the naive case, we can now have pebbles of differing masses, and we can also have a countably infinite number of pebbles as long as their total mass is 1.

We can even have uncountable sample spaces, such as having S be an area in the plane. In this case, instead of pebbles, we can visualize mud spread out over a region, where the total mass of the mud is 1.

Any function P (mapping events to numbers in the interval $[0, 1]$) that satisfies the two axioms is considered a valid probability function. However, the axioms don't tell us how probability should be *interpreted*; different schools of thought exist.

The *frequentist* view of probability is that it represents a long-run frequency over a large number of repetitions of an experiment: if we say a coin has probability $1/2$ of Heads, that means the coin would land Heads 50% of the time if we tossed it over and over and over.

The *Bayesian* view of probability is that it represents a degree of belief about the event in question, so we can assign probabilities to hypotheses like "candidate A will win the election" or "the defendant is guilty" even if it isn't possible to repeat the same election or the same crime over and over again.

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The Bayesian and frequentist perspectives are complementary, and both will be helpful for developing intuition in later chapters. Regardless of how we choose to interpret probability, we can use the two axioms to derive other properties of prob

ability, and these results will hold for *any* valid probability function.

Theorem 1.6.2 (Properties of probability). Probability has the following properties, for any events A and B .

1. $P(A^c) = 1 - P(A)$.
2. If $A \subseteq B$, then $P(A) \leq P(B)$.
3. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

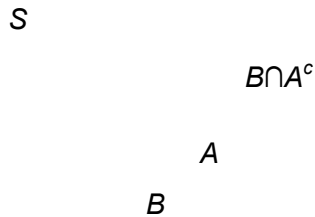
Proof.

1. Since A and A^c are disjoint and their union is S , the second axiom gives

$$P(S) = P(A \cup A^c) = P(A) + P(A^c),$$

But $P(S) = 1$ by the first axiom. So $P(A) + P(A^c) = 1$.

2. If $A \subseteq B$, then we can write B as the union of A and $B \cap A^c$, where $B \cap A^c$ is the part of B not also in A . This is illustrated in the Venn diagram below.

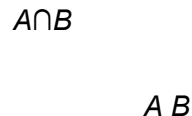


Since A and $B \cap A^c$ are disjoint, we can apply the second

$$\text{axiom: } P(B) = P(A \cup (B \cap A^c)) = P(A) + P(B \cap A^c).$$

Probability is nonnegative, so $P(B \cap A^c) \geq 0$, proving that $P(B) \geq P(A)$. 3. The

intuition for this result can be seen using a Venn diagram like the one below. S



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The shaded region represents $A \cup B$, but the probability of this region is not the sum $P(A) + P(B)$, because that would count the football-shaped region $A \cap B$ twice. To correct for this, we subtract $P(A \cap B)$. This is a useful intuition, but not a proof.

For a proof using the axioms of probability, we can write $A \cup B$ as the union of the disjoint events A and $B \cap A^c$. Then by the second axiom,

$$P(A \cup B) = P(A \cup (B \cap A^c)) = P(A) + P(B \cap A^c).$$

So it suffices to show that $P(B \cap A^c) = P(B) - P(A \cap B)$. Since $A \cap B$ and $B \cap A^c$ are disjoint and their union is B , another application of the second axiom gives us

$$P(A \cap B) + P(B \cap A^c) = P(B).$$

So $P(B \cap A^c) = P(B) - P(A \cap B)$, as desired.

The third property is a special case of *inclusion-exclusion*, a formula for finding the probability of a union of events when the events are not necessarily disjoint. We showed above that for two events A and B ,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

For three events, inclusion-exclusion says

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) \\ &\quad - P(A \cap B) - P(A \cap C) - P(B \cap C) \\ &\quad + P(A \cap B \cap C). \end{aligned}$$

For intuition, consider a triple Venn diagram like the one below.

$A \ B$

C

To get the total area of the shaded region $A \cup B \cup C$, we start by adding the areas of the three circles, $P(A) + P(B) + P(C)$. The three football-shaped regions have each been counted twice, so we then subtract $P(A \cap B) + P(A \cap C) + P(B \cap C)$. Finally, the region in the center has been added three times and subtracted three times, so in order to count it exactly once, we must add it back again. This ensures that each region of the diagram has been counted once and exactly once.

Now we can write inclusion-exclusion for n events.

Probability and counting 25 Theorem 1.6.3 (Inclusion-exclusion). For any events $A_1,$

$\dots, A_n,$

$$\begin{aligned} P(A_1 \cup \dots \cup A_n) &= \sum_{i=1}^n P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \sum_{i < j < k} P(A_i \cap A_j \cap A_k) - \dots + (-1)^{n+1} P(A_1 \cap \dots \cap A_n). \end{aligned}$$

This formula can be proven by induction using just the axioms, but instead we'll present a shorter proof in Chapter 4 after introducing some additional tools. The rationale behind the alternating addition and subtraction in the general formula is analogous to the special cases we've already considered.

The next example, *de Montmort's matching problem*, is a famous application of inclusion-exclusion. Pierre R'emonde de Montmort was a French mathematician who studied probability in the context of gambling and wrote a treatise [19] devoted to the analysis of various card games. He posed the following problem in 1708, based on a card game called Treize.

Example 1.6.4 (de Montmort's matching problem). Consider a well-shuffled deck of n cards, labeled 1 through n . You flip over the cards one by one, saying the numbers 1 through n as you do so. You win the game if, at some point, the number you say aloud is the same as the number on the card being flipped over (for example, if the 7th card in the deck has the label 7). What is the probability of winning?

Solution:

Let A_i be the event that the i th card in the deck has the number i written on it. We are interested in the probability of the union $A_1 \cup \cdots \cup A_n$: as long as at least one of the cards has a number matching its position in the deck, you will win the game. (An ordering for which you lose is called a *derangement*, though hopefully no one has ever become deranged due to losing at this game.)

To find the probability of the union, we'll use inclusion-exclusion. First,

$$P(A_i) = \frac{1}{n}$$

for all i . One way to see this is with the naive definition of probability, using the full sample space: there are $n!$ possible orderings of the deck, all equally likely, and $(n-1)!$ of these are favorable to A_i (fix the card numbered i to be in the i th position in the deck, and then the remaining $n - 1$ cards can be in any order). Another way to see this is by symmetry: the card numbered i is equally likely to be in any of the n positions in the deck, so it has probability $1/n$ of being in the i th spot. Second,

$$P(A_i \cap A_j) = \frac{1}{n(n-1)}$$

$$P(A_i \cap A_j \cap A_k) = \frac{1}{n(n-1)(n-2)}$$

$$P(A_i \cap A_j \cap A_k \cap A_l) = \frac{1}{n(n-1)(n-2)(n-3)}$$

since we require the cards numbered i and j to be in the i th and j th spots in the deck and allow the remaining $n - 2$ cards to be in any order, so $(n - 2)!$ out of $n!$ possibilities are favorable to $A_i \cap A_j$. Similarly,

$$P(A_i \cap A_j \cap A_k) = \frac{1}{n(n-1)(n-2)}$$

$$P(A_i \cap A_j \cap A_k \cap A_l) = \frac{1}{n(n-1)(n-2)(n-3)}$$

and the pattern continues for intersections of 4 events, etc.

In the inclusion-exclusion formula, there are n terms involving one event, $\binom{n}{2}$ terms involving two events, $\binom{n}{3}$ terms involving three events, and so forth. By the symmetry of the problem, all n terms of the form $P(A_i)$ are equal, all $\binom{n}{2}$ terms of the

form $P(A_i \cap A_j)$ are equal, and the whole expression simplifies considerably:

$$\begin{aligned}
 P_{i=1}^n A_i &= \sum_{i=1}^n P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \sum_{i < j < k} P(A_i \cap A_j \cap A_k) - \cdots + (-1)^{n+1} \cdot 1/n! \\
 &= n \cdot \frac{1}{n} - \frac{n(n-1)}{2} \cdot \frac{1}{n(n-1)} + \frac{n(n-1)(n-2)}{6} \cdot \frac{1}{n(n-1)(n-2)} - \cdots + (-1)^{n+1} \cdot \frac{1}{n!} \\
 &= 1 - \frac{1}{2!} + \frac{1}{3!} - \cdots + (-1)^{n+1} \cdot \frac{1}{n!}.
 \end{aligned}$$

Comparing this to the Taylor series for $1/e$ (see Section A.8 of the math

$$\text{appendix}), e^{-1} = 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots,$$

we see that for large n , the probability of winning the game is extremely close to $1 - 1/e$, or about 0.63. Interestingly, as n grows, the probability of winning approaches $1 - 1/e$ instead of going to 0 or 1. With a lot of cards in the deck, the number of possible locations for matching cards increases while the probability of any particular match decreases, and these two forces offset each other and balance to give a probability of about $1 - 1/e$.

Inclusion-exclusion is a very general formula for the probability of a union of events, but it helps us the most when there is symmetry among the events A_j ; otherwise the sum can be extremely tedious. In general, when symmetry is lacking, we should try to use other tools before turning to inclusion-exclusion as a last resort.

1.7 Recap

Probability allows us to quantify uncertainty and randomness in a principled way. Probabilities arise when we perform an experiment: the set of all possible outcomes of the experiment is called the sample space, and a subset of the sample space is called an event. It is useful to be able to go back and forth between describing events in English and writing them down mathematically as sets (often using unions, intersections, and complements).

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Pebble World can help us visualize sample spaces and events when the sample space is finite. In Pebble World, each outcome is a pebble, and an event is a set of pebbles. If all the pebbles have the same mass (i.e., are equally likely), we can apply the naive definition of probability, which lets us calculate probabilities by counting.

To this end, we discussed several tools for counting. When counting the number of possibilities, we often use the multiplication rule. For example, there are $n!$ permutations of the numbers $1, 2, \dots, n$ and there are 2^n subsets of a set with n elements. If we can't directly use the multiplication rule, we can sometimes count each possibility exactly c times for some c , and then divide by c to get the actual number of possibilities. For example, this strategy is useful for finding an expression for binomial coefficients in terms of factorials.

An important pitfall to avoid is misapplying the naive definition of probability, implicitly or explicitly assuming equally likely outcomes without justification. One technique to help avoid this is to *give objects labels*, for precision and so that we are not tempted to treat them as indistinguishable.

Moving beyond the naive definition, we define probability to be a function that takes an event and assigns to it a real number between 0 and 1. We require a valid probability function to satisfy two axioms:

1. $P(\emptyset) = 0$, $P(S) = 1$.
2. If $A_1, A_2, \dots$ are disjoint events, then

$$P\left(\bigcup_{j=1}^{\infty} A_j\right) = \sum_{j=1}^{\infty} P(A_j).$$

Many useful properties can be derived from these axioms. For

$$\text{example, } P(A^c) = 1 - P(A)$$

for any event A , and we have the inclusion-exclusion formula

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \sum_{i < j < k} P(A_i \cap A_j \cap A_k) - \dots + (-1)^{n+1} P(A_1 \cap \dots \cap A_n)$$

for any events $A_1, \dots, A_n$. For $n = 2$, this is the much nicer-looking result

$$P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2).$$

Figure 1.7 illustrates how a probability function maps events to numbers between 0 and 1. We'll add many new concepts to this diagram as we continue our journey through the field of probability.

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What can
happen?

		events numbers
A		$P(A^c) = 1 - P(A)$
not A	FIGURE 1.7	$P(A \cap B)$
A and B		$P(A \cup B) = P(A) + P(B) -$
A or B	P	$P(A \cap B) \quad P(S) = 1$
something happened	$P(A)$	

It is important to distinguish between *events* and *probabilities*. The former are sets, while the latter are numbers. Before the experiment is done, we generally don't know whether or not a particular event will occur (happen). So we assign it a probability of happening, using a probability function P . We can use set operations to define new events in terms of old events, and the properties of probabilities to relate the probabilities of the new events to those of the old events.

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1.8 R

R is a very powerful, popular environment for statistical computing and graphics, freely available for Mac OS X, Windows, and UNIX systems. Knowing how to use R is an extremely useful skill. R and various supporting information can be obtained at <https://www.r-project.org>. RStudio is an excellent alternative interface for R, freely available at <https://www.rstudio.com>.

In the R section at the end of each chapter, we provide R code to let you try out some of the examples from the chapter, especially via simulation. These sections are not intended to be a full introduction to R; many R tutorials are available for free online, and many books on R are available. But the R sections show how to implement various simulations, computations, and visualizations that naturally accompany the material from each chapter. The R code at the end of each chapter is also available at <http://stat110.net>.

Vectors

R is built around *vectors*, and getting familiar with “vectorized thinking” is very important for using R effectively. To create a vector, we can use the `c` command (which stands for *combine* or *concatenate*). For example,

```
v <- c(3,1,4,1,5,9)
```

defines `v` to be the vector (3, 1, 4, 1, 5, 9). (The left arrow `<-` is typed as `<` followed by `-`. The symbol `=` can be used instead, but the arrow is more suggestive of the fact that the variable on the left is being set equal to the value on the right.) Similarly, `n <- 110` sets `n` equal to 110; R views `n` as a vector of length 1.

```
sum(v)
```

adds up the entries of v , $\max(v)$ gives the largest value, $\min(v)$ gives the smallest value, and $\text{length}(v)$ gives the length.

A shortcut for getting the vector $(1, 2, \dots, n)$ is to type $1:n$; more generally, if m and n are integers then $m:n$ gives the sequence of integers from m to n (in increasing order if $m \leq n$ and in decreasing order otherwise).

To access the i th entry of a vector v , use $v[i]$. We can also get subvectors easily:

```
v[c(1,3,5)]
```

gives the vector consisting of the 1st, 3rd, and 5th entries of v . It's also possible to get a subvector by specifying what to exclude, using a minus sign:

```
v[-(2:4)]
```

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gives the vector obtained by removing the 2nd through 4th entries of v (the parentheses are needed since $-2:4$ would be $(-2, -1, \dots, 4)$).

Many operations in R are interpreted *componentwise*. For example, in math the cube of a vector doesn't have a standard definition, but in R typing v^3 simply cubes each entry individually. Similarly,

```
1/(1:100)^2
```

is a very compact way to get the vector $(1, \frac{1}{2^2}, \frac{1}{3^2}, \dots, \frac{1}{100^2})$.

In math, $v+w$ is undefined if v and w are vectors of different lengths, but in R the shorter vector gets "recycled"! For example, $v+3$ adds 3 to each entry of v .

Factorials and binomial coefficients

We can compute $n!$ using $\text{factorial}(n)$ and $\binom{n}{k}$ using $\text{choose}(n,k)$. As we have seen, factorials grow extremely quickly. What is the largest n for which R returns a number for $\text{factorial}(n)$? Beyond that point, R will return Inf (infinity), with a warning message. But it may still be possible to use $\text{lfactorial}(n)$ for larger values of n , which computes $\log(n!)$. Similarly, $\text{lchoose}(n,k)$ computes $\log \binom{n}{k}$.

Sampling and simulation

The `sample` command is a useful way of drawing random samples in R. (Technically, they are *pseudo-random* since there is an underlying deterministic algorithm, but they "look like" random samples for almost all practical purposes.) For example,

```
n <- 10; k <- 5
sample(n,k)
```

generates an ordered random sample of 5 of the numbers from 1 to 10, without replacement, and with equal probabilities given to each number. To sample with replacement instead, just add in `replace = TRUE`:

```
n <- 10; k <- 5
```

```
sample(n,k,replace=TRUE)
```

To generate a random permutation of $1, 2, \dots, n$ we can use `sample(n,n)`, which because of R's default settings can be abbreviated to `sample(n)`.

We can also use `sample` to draw from a non-numeric vector. For example, `letters` is built into R as the vector consisting of the 26 lowercase letters of the English alphabet, and `sample(letters,7)` will generate a random 7-letter “word” by sampling from the alphabet, without replacement.

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The `sample` command also allows us to specify general probabilities for sampling each number. For example,

```
sample(4, 3, replace=TRUE, prob=c(0.1,0.2,0.3,0.4))
```

samples three numbers between 1 and 4, with replacement, and with probabilities given by $(0.1, 0.2, 0.3, 0.4)$. If the sampling is without replacement, then at each stage the probability of any not-yet-chosen number is *proportional* to its original probability.

Generating *many* random samples allows us to perform a *simulation* for a probability problem. The `replicate` command, which is explained below, is a convenient way to do this.

Matching problem simulation

Let's show by simulation that the probability of a matching card in Example 1.6.4 is approximately $1 - 1/e$ when the deck is sufficiently large. Using R, we can perform the experiment a bunch of times and see how many times we encounter at least one matching card:

```
n <- 100
r <- replicate(10^4, sum(sample(n)==(1:n)))
sum(r>=1)/10^4
```

In the first line, we choose how many cards are in the deck (here, 100 cards). In the second line, let's work from the inside out:

- `sample(n)==(1:n)` is a vector of length n , the i th element of which equals 1 if the i th card matches its position in the deck and 0 otherwise. That's because for two numbers a and b , the expression `a==b` is TRUE if $a = b$ and FALSE otherwise, and TRUE is encoded as 1 and FALSE is encoded as 0.
- `sum` adds up the elements of the vector, giving us the number of matching cards in this run of the experiment.
 - `replicate` does this 10^4 times. We store the results in `r`, a vector of length 10^4 containing the numbers of matched cards from each run of the experiment.

In the last line, we add up the number of times where there was at least one matching card, and we divide by the number of simulations.

To explain what the code is doing within the code rather than in separate documentation, we can add *comments* using the `#` symbol to mark the start of a comment. Comments are ignored by R but can make the code much easier to understand for the reader (who could be you—even if you will be the only one

using your code, it is often hard to remember what everything means and how the code is supposed to work when looking at it a month after writing it). Short comments can be on the
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same line as the corresponding code; longer comments should be on separate lines. For example, a commented version of the above simulation is:

```
n <- 100 # number of cards r <- replicate(10^4, sum(sample(n)==(1:n))) # shuffle;
count matches sum(r>=1)/10^4 # proportion with a match
```

What did you get when you ran the code? We got 0.63, which is quite close to $1 - 1/e$.

Birthday problem calculation and simulation

The following code uses `prod` (which gives the product of a vector) to calculate the probability of at least one birthday match in a group of 23 people:

```
k <- 23
1-prod((365-k+1):365)/365^k
```

Better yet, R has built-in functions, `pbirthday` and `qbirthday`, for the birthday problem! `pbirthday(k)` returns the probability of at least one match if the room has k people. `qbirthday(p)` returns the number of people needed in order to have probability p of at least one match. For example, `pbirthday(23)` is 0.507 and `qbirthday(0.5)` is 23.

We can also find the probability of having at least one *triple birthday match*, i.e., three people with the same birthday; all we have to do is add `coincident=3` to say we're looking for triple matches. For example, `pbirthday(23,coincident=3)` returns 0.014, so 23 people give us only a 1.4% chance of a triple birthday match. `qbirthday(0.5,coincident=3)` returns 88, so we'd need 88 people to have at least a 50% chance of at least one triple birthday match.

To simulate the birthday problem, we can use

```
b <- sample(1:365,23,replace=TRUE)
tabulate(b)
```

to generate random birthdays for 23 people and then tabulate the counts of how many people were born on each day (the command `table(b)` creates a prettier table, but is slower). We can run 10^4 repetitions as follows:

```
r <- replicate(10^4, max(tabulate(sample(1:365,23,replace=TRUE))))
sum(r>=2)/10^4
```

If the probabilities of various days are not all equal, the calculation becomes much more difficult, but the simulation can easily be extended since `sample` allows us to specify the probability of each day (by default `sample` assigns equal probabilities, so in the above the probability is $1/365$ for each day).

1.9 Exercises

Exercises marked with s have detailed solutions at <http://stat110.net>.

Counting

1. How many ways are there to permute the letters in the word MISSISSIPPI?
2. (a) How many 7-digit phone numbers are possible, assuming that the first digit can't be a 0 or a 1?

(b) Re-solve (a), except now assume also that the phone number is not allowed to start with 911 (since this is reserved for emergency use, and it would not be desirable for the system to wait to see whether more digits were going to be dialed after someone has dialed 911).
3. Fred is planning to go out to dinner each night of a certain week, Monday through Friday, with each dinner being at one of his ten favorite restaurants.

(a) How many possibilities are there for Fred's schedule of dinners for that Monday through Friday, if Fred is not willing to eat at the same restaurant more than once?

(b) How many possibilities are there for Fred's schedule of dinners for that Monday through Friday, if Fred is willing to eat at the same restaurant more than once, but is not willing to eat at the same place twice in a row (or more)?
4. A *round-robin tournament* is being held with n tennis players; this means that every player will play against every other player exactly once.

(a) How many possible outcomes are there for the tournament (the outcome lists out who won and who lost for each game)?

(b) How many games are played in total?
5. A *knock-out tournament* is being held with 2^n tennis players. This means that for each round, the winners move on to the next round and the losers are eliminated, until only one person remains. For example, if initially there are $2^4 = 16$ players, then there are 8 games in the first round, then the 8 winners move on to round 2, then the 4 winners move on to round 3, then the 2 winners move on to round 4, the winner of which is declared the winner of the tournament. (There are various systems for determining who plays whom within a round, but these do not matter for this problem.)

(a) How many rounds are there?

(b) Count how many games in total are played, by adding up the numbers of games played in each round.

(c) Count how many games in total are played, this time by directly thinking about it without doing almost any calculation.

Hint: How many players need to be eliminated?
6. There are 20 people at a chess club on a certain day. They each find opponents and start playing. How many possibilities are there for how they are matched up, assuming that in each game it *does* matter who has the white pieces (in a chess game, one player has the white pieces and the other player has the black pieces)?
7. Two chess players, A and B, are going to play 7 games. Each game has three possible outcomes: a win for A (which is a loss for B), a draw (tie), and a loss for A (which is a win for B). A win is worth 1 point, a draw is worth 0.5 points, and a loss is worth 0 points.

(a) How many possible outcomes for the individual games are there, such that overall player A ends up with 3 wins, 2 draws, and 2 losses?

(b) How many possible outcomes for the individual games are there, such that A ends

up with 4 points and B ends up with 3 points?

(c) Now assume that they are playing a best-of-7 match, where the match will end when either player has 4 points or when 7 games have been played, whichever is first. For example, if after 6 games the score is 4 to 2 in favor of A, then A wins the match and they don't play a 7th game. How many possible outcomes for the individual games are there, such that the match lasts for 7 games and A wins by a score of 4 to 3?

8. s (a) How many ways are there to split a dozen people into 3 teams, where one team has 2 people, and the other two teams have 5 people each?

(b) How many ways are there to split a dozen people into 3 teams, where each team has 4 people?

9. s (a) How many paths are there from the point (0, 0) to the point (110, 111) in the plane such that each step either consists of going one unit up or one unit to the right?

(b) How many paths are there from (0, 0) to (210, 211), where each step consists of going one unit up or one unit to the right, and the path has to go through (110, 111)? 10. To fulfill the requirements for a certain degree, a student can choose to take any 7 out of a list of 20 courses, with the constraint that at least 1 of the 7 courses must be a statistics course. Suppose that 5 of the 20 courses are statistics courses.

(a) How many choices are there for which 7 courses to take?

(b) Explain intuitively why the answer to (a) is *not* $\binom{5}{1} \cdot \binom{19}{6}$.

11. Let A and B be sets with $|A| = n$, $|B| = m$.

(a) How many functions are there from A to B (i.e., functions with domain A , assigning an element of B to each element of A)?

(b) How many one-to-one functions are there from A to B ? (See Section A.2.1 of the math appendix for information about one-to-one functions.)

12. Four players, named A, B, C, and D, are playing a card game. A standard, well-shuffled deck of cards is dealt to the players (so each player receives a 13-card hand).

(a) How many possibilities are there for the hand that player A will get? (Within a hand, the order in which cards were received doesn't matter.)

(b) How many possibilities are there overall for what hands everyone will get, assuming that it matters which player gets which hand, but not the order of cards within a hand?

(c) Explain intuitively why the answer to Part (b) is not the fourth power of the answer to Part (a).

13. A certain casino uses 10 standard decks of cards mixed together into one big deck, which we will call a *superdeck*. Thus, the superdeck has $52 \cdot 10 = 520$ cards, with 10 copies of each card. How many different 10-card hands can be dealt from the superdeck? The order of the cards does not matter, nor does it matter which of the original 10 decks the cards came from. Express your answer as a binomial coefficient.

Hint: Bose-Einstein.

14. You are ordering two pizzas. A pizza can be small, medium, large, or extra large, with any combination of 8 possible toppings (getting no toppings is allowed, as is getting all 8). How many possibilities are there for your two pizzas?

Probability and counting 35 Story proofs

15. s Give a story proof that $\sum_{k=0}^n \binom{n}{k} = 2^n$.

16. s Show that for all positive integers n and k with $n \geq k$,

$$\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k}$$

doing this in two ways: (a) algebraically and (b) with a story, giving an interpretation for why both sides count the same thing.

Hint for the story proof: Imagine an organization consisting of $n + 1$ people, with one of them pre-designated as the president of the organization.

17. Give a story proof that

integers n . story proof
for all positive 18. Give a that
, n

$$\sum_{k=0}^n \binom{n}{k}^2 = \sum_{k=0}^n \binom{n}{k} \binom{n}{n-k} = \sum_{k=0}^n \binom{n}{k} = 2^n$$

positive integers n , for all n

Hint: Consider choosing a committee of size n from two groups of size n each, where only one of the two groups has people eligible to become the chair of the committee.

19. Give a story proof that

$$\sum_{k=2}^n (n-k+2) = \frac{n(n+3)}{2}$$

for all integers $n \geq 2$

Hint: Consider the middle number in a subset of $\{1, 2, \dots, n+3\}$ of size 5.

20. s (a) Show using a story proof that

$$\binom{n}{k} + \binom{n}{k+1} + \binom{n}{k+2} + \dots + \binom{n}{n} = \binom{n+1}{k+1}$$

where n and k are positive integers with $n \geq k$. This is called the *hockey stick identity*. Hint: Imagine arranging a group of people by age, and then think about the oldest person in a chosen subgroup.

(b) Suppose that a large pack of Haribo gummi bears can have anywhere between 30 and 50 gummi bears. There are 5 delicious flavors: pineapple (clear), raspberry (red), orange (orange), strawberry (green, mysteriously), and lemon (yellow). There are 0 non delicious flavors. How many possibilities are there for the composition of such a pack of gummi bears? You can leave your answer in terms of a couple binomial coefficients, but not a sum of lots of binomial coefficients.

21. Define n_k as the number of ways to partition $\{1, 2, \dots, n\}$ into k nonempty subsets, or the number of ways to have n students split up into k groups such that each group has at least one student. For example, $4_2 = 7$ because we have the following possibilities.

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- $\{1\}, \{2, 3, 4\}$
- $\{2\}, \{1, 3, 4\}$
- $\{3\}, \{1, 2, 4\}$
- $\{4\}, \{1, 2, 3\}$
- $\{1, 2\}, \{3, 4\}$
- $\{1, 3\}, \{2, 4\}$
- $\{1, 4\}, \{2, 3\}$

Prove the following identities: (a)

$$\sum_{k=1}^n \binom{n}{k} k = n \cdot 2^{n-1}$$

Hint: I'm either in a group by myself or I'm not.

(b)

$$\sum_{j=k}^n \binom{n}{j} j = \sum_{j=k}^n \binom{n}{j} j = n \cdot 2^{n-1}$$

Hint: First decide how many people are not going to be in my group.

22. The Dutch mathematician R.J. Stroeker remarked:

Every beginning student of number theory surely must have marveled at the miraculous fact that for each natural number n the sum of the first n positive consecutive cubes is a perfect square. [26]

Furthermore, it is the square of the sum of the first n positive integers! That is, 1^3

$$+ 2^3 + \cdots + n^3 = (1 + 2 + \cdots + n)^2.$$

Usually this identity is proven by induction, but that does not give much insight into why the result is true, nor does it help much if we wanted to compute the left-hand side but didn't already know this result. In this problem, you will give a story proof of the identity.

(a) Give a story proof of the identity

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Hint: Consider a round-robin tournament (see Exercise 4).

(b) Give a story proof of the identity

$$1^3 + 2^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}.$$

It is then just basic algebra (not required for this problem) to check that the square of the right-hand side in (a) is the right-hand side in (b).

Hint: Imagine choosing a number between 1 and n and then choosing 3 numbers between 0 and n smaller than the original number, with replacement. Then consider cases based on how many distinct numbers were chosen.

Naive definition of probability

23. Three people get into an empty elevator at the first floor of a building that has 10 floors. Each presses the button for their desired floor (unless one of the others has already pressed that button). Assume that they are equally likely to want to go to floors 2 through 10 (independently of each other). What is the probability that the buttons for 3 consecutive floors are pressed?

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24. s A certain family has 6 children, consisting of 3 boys and 3 girls. Assuming that all birth orders are equally likely, what is the probability that the 3 eldest children are the 3 girls?
25. s A city with 6 districts has 6 robberies in a particular week. Assume the robberies are located randomly, with all possibilities for which robbery occurred where equally likely. What is the probability that some district had more than 1 robbery?
26. A survey is being conducted in a city with 1 million residents. It would be far too expensive to survey all of the residents, so a random sample of size 1000 is chosen (in practice, there are many challenges with sampling, such as obtaining a complete list of everyone in the city, and dealing with people who refuse to participate). The survey is conducted by choosing people one at a time, *with* replacement and with equal probabilities.
- (a) Explain how sampling with vs. without replacement here relates to the birthday problem.
- (b) Find the probability that at least one person will get chosen more than once.
27. A *hash table* is a commonly used data structure in computer science, allowing for fast information retrieval. For example, suppose we want to store some people's phone numbers. Assume that no two of the people have the same name. For each name x , a *hash function* h is used, letting $h(x)$ be the location that will be used to store x 's phone number. After such a table has been computed, to look up x 's phone number one just recomputes $h(x)$ and then looks up what is stored in that location.

The hash function h is deterministic, since we don't want to get different results every time we compute $h(x)$. But h is often chosen to be *pseudorandom*. For this problem,

assume that true randomness is used. Let there be k people, with each person's phone number stored in a random location (with equal probabilities for each location, independently of where the other people's numbers are stored), represented by an integer between 1 and n . Find the probability that at least one location has more than one phone number stored there.

28. s A college has 10 time slots for its courses, and blithely assigns courses to completely random time slots, independently. The college offers exactly 3 statistics courses. What is the probability that 2 or more of the statistics courses are in the same time slot?
29. s For each part, decide whether the blank should be filled in with $=$, $<$, or $>$, and give a clear explanation.
- (a) (probability that the total after rolling 4 fair dice is 21) (probability that the total after rolling 4 fair dice is 22)
- (b) (probability that a random 2-letter word is a palindrome³) (probability that a random 3-letter word is a palindrome)
30. With definitions as in the previous problem, find the probability that a random n -letter word is a palindrome for $n = 7$ and for $n = 8$.
31. s Elk dwell in a certain forest. There are N elk, of which a simple random sample of size n are captured and tagged ("simple random sample" means that all $\binom{N}{n}$ sets of n elk are equally likely). The captured elk are returned to the population, and then a new sample is drawn, this time with size m . This is an important method that is widely used in ecology, known as *capture-recapture*. What is the probability that exactly k of the m

³A *palindrome* is an expression such as "A man, a plan, a canal: Panama" that reads the same backwards as forwards (ignoring spaces, capitalization, and punctuation). Assume for this problem that all words of the specified length are equally likely, that there are no spaces or punctuation, and that the alphabet consists of the lowercase letters a, b, . . . , z. A word is any string of letters from the alphabet; it does not need to be a word that has a meaning in the English language.

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elk in the new sample were previously tagged? (Assume that an elk that was captured before doesn't become more or less likely to be captured again.)

32. Four cards are face down on a table. You are told that two are red and two are black, and you need to guess which two are red and which two are black. You do this by pointing to the two cards you're guessing are red (and then implicitly you're guessing that the other two are black). Assume that all configurations are equally likely, and that you do not have psychic powers. Find the probability that exactly j of your guesses are correct, for $j = 0, 1, 2, 3, 4$.
33. s A jar contains r red balls and g green balls, where r and g are fixed positive integers. A ball is drawn from the jar randomly (with all possibilities equally likely), and then a second ball is drawn randomly.
- (a) Explain intuitively why the probability of the second ball being green is the same as the probability of the first ball being green.
- (b) Define notation for the sample space of the problem, and use this to compute the probabilities from (a) and show that they are the same.
- (c) Suppose that there are 16 balls in total, and that the probability that the two balls are the same color is the same as the probability that they are different colors. What are r and g (list all possibilities)?
34. s A random 5-card poker hand is dealt from a standard deck of cards. Find the probability of each of the following possibilities (in terms of binomial coefficients).
- (a) A flush (all 5 cards being of the same suit; do not count a royal flush, which is a flush with an ace, king, queen, jack, and 10).
- (b) Two pair (e.g., two 3's, two 7's, and an ace).
35. A random 13-card hand is dealt from a standard deck of cards. What is the probability that the hand contains at least 3 cards of every suit?
36. A group of 30 dice are thrown. What is the probability that 5 of each of the values 1, 2, 3, 4, 5, 6 appear?

37. A deck of cards is shuffled well. The cards are dealt one by one, until the first time an ace appears.
- (a) Find the probability that no kings, queens, or jacks appear before the first ace.
 - (b) Find the probability that exactly one king, exactly one queen, and exactly one jack appear (in any order) before the first ace.
38. Tyrion, Cersei, and ten other people are sitting at a round table, with their seating arrangement having been randomly assigned. What is the probability that Tyrion and Cersei are sitting next to each other? Find this in two ways:
- (a) using a sample space of size $12!$, where an outcome is fully detailed about the seating; (b) using a much smaller sample space, which focuses on Tyrion and Cersei.
39. An organization with $2n$ people consists of n married couples. A committee of size k is selected, with all possibilities equally likely. Find the probability that there are exactly j married couples within the committee.
40. There are n balls in a jar, labeled with the numbers $1, 2, \dots, n$. A total of k balls are drawn, one by one *with replacement*, to obtain a sequence of numbers.
- (a) What is the probability that the sequence obtained is strictly increasing?
 - (b) What is the probability that the sequence obtained is increasing (but not necessarily strictly increasing, i.e., there can be repetitions)?

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41. Each of n balls is independently placed into one of n boxes, with all boxes equally likely. What is the probability that exactly one box is empty?
42. A *norepeatword* is a sequence of at least one (and possibly all) of the usual 26 letters a,b,c,...,z, with repetitions not allowed. For example, "course" is a norepeatword, but "statistics" is not. Order matters, e.g., "course" is not the same as "source". A norepeatword is chosen randomly, with all norepeatwords equally likely. Show that the probability that it uses all 26 letters is very close to $1/e$.

Axioms of probability

43. Show that for any events A and B ,

$$P(A) + P(B) - 1 \leq P(A \cap B) \leq P(A \cup B) \leq P(A) + P(B).$$

For each of these three inequalities, give a simple criterion for when the inequality is actually an equality (e.g., give a simple condition such that $P(A \cap B) = P(A \cup B)$ if and only if the condition holds).

44. Let A and B be events. The *difference* $B - A$ is defined to be the set of all elements of B that are not in A . Show that if $A \subseteq B$, then

$$P(B - A) = P(B) - P(A),$$

directly using the axioms of probability.

45. Let A and B be events. The *symmetric difference* AB is defined to be the set of all elements that are in A or B but not both. In logic and engineering, this event is also called the *XOR (exclusive or)* of A and B . Show that

$$P(AB) = P(A) + P(B) - 2P(A \cap B),$$

directly using the axioms of probability.

46. Let $A_1, A_2, \dots, A_n$ be events. Let B_k be the event exactly k of the A_i occur, and C_k be the event that at least k of the A_i occur, for $0 \leq k \leq n$. Find a simple expression for $P(B_k)$ in terms of $P(C_k)$ and $P(C_{k+1})$.

47. Events A and B are *independent* if $P(A \cap B) = P(A)P(B)$ (independence is explored in detail in the next chapter).

- (a) Give an example of independent events A and B in a finite sample space S (with

neither equal to $\emptyset$ or S), and illustrate it with a Pebble World diagram.

(b) Consider the experiment of picking a random point in the rectangle

$$R = \{(x, y) : 0 < x < 1, 0 < y < 1\},$$

where the probability of the point being in any particular region contained within R is the area of that region. Let A_1 and B_1 be rectangles contained within R , with areas not equal to 0 or 1. Let A be the event that the random point is in A_1 , and B be the event that the random point is in B_1 . Give a geometric description of when it is true that A and B are independent. Also, give an example where they are independent and another example where they are not independent.

(c) Show that if A and B are independent, then

$$P(A \cup B) = P(A) + P(B) - P(A)P(B) = 1 - P(A^c)P(B^c).$$

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48. s Arby has a belief system assigning a number $P_{\text{Arby}}(A)$ between 0 and 1 to every event A (for some sample space). This represents Arby's degree of belief about how likely A is to occur. For any event A , Arby is willing to pay a price of $1000 \cdot P_{\text{Arby}}(A)$ dollars to buy a certificate such as the one shown below:

Certificate

The owner of this certificate can redeem it for \$1000 if A occurs. No value if A does not occur, except as required by federal, state, or local law. No expiration date.

Likewise, Arby is willing to sell such a certificate at the same price. Indeed, Arby is willing to buy or sell any number of certificates at this price, as Arby considers it the "fair" price.

Arby stubbornly refuses to accept the axioms of probability. In particular, suppose that there are two *disjoint* events A and B with

$$P_{\text{Arby}}(A \cup B) = P_{\text{Arby}}(A) + P_{\text{Arby}}(B).$$

Show how to make Arby go bankrupt, by giving a list of transactions Arby is willing to make that will *guarantee* that Arby will lose money (you can assume it will be known whether A occurred and whether B occurred the day after any certificates are bought/sold).

Inclusion-exclusion

49. A fair die is rolled n times. What is the probability that at least 1 of the 6 values never appears?
50. s A card player is dealt a 13-card hand from a well-shuffled, standard deck of cards. What is the probability that the hand is void in at least one suit ("void in a suit" means having no cards of that suit)?
51. s For a group of 7 people, find the probability that all 4 seasons (winter, spring, summer, fall) occur at least once each among their birthdays, assuming that all seasons are equally likely.
52. A certain class has 20 students, and meets on Mondays and Wednesdays in a classroom with exactly 20 seats. In a certain week, everyone in the class attends both days. On both days, the students choose their seats completely randomly (with one student per seat). Find the probability that no one sits in the same seat on both days of that week.
53. Fred needs to choose a password for a certain website. Assume that he will choose an 8-character password, and that the legal characters are the lowercase letters a, b, c, . . . , z, the uppercase letters A, B, C, . . . , Z, and the numbers 0, 1, . . . , 9.

(a) How many possibilities are there if he is required to have at least one lowercase

letter in his password?

(b) How many possibilities are there if he is required to have at least one lowercase letter and at least one uppercase letter in his password?

(c) How many possibilities are there if he is required to have at least one lowercase letter, at least one uppercase letter, and at least one number in his password?

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54. s Alice attends a small college in which each class meets only once a week. She is deciding between 30 non-overlapping classes. There are 6 classes to choose from for each day of the week, Monday through Friday. Trusting in the benevolence of randomness, Alice decides to register for 7 randomly selected classes out of the 30, with all choices equally likely. What is the probability that she will have classes every day, Monday through Friday? (This problem can be done either directly using the naive definition of probability, or using inclusion-exclusion.)

55. A club consists of 10 seniors, 12 juniors, and 15 sophomores. An organizing committee of size 5 is chosen randomly (with all subsets of size 5 equally likely).

(a) Find the probability that there are exactly 3 sophomores in the committee.

(b) Find the probability that the committee has at least one representative from each of the senior, junior, and sophomore classes.

Mixed practice

56. For each part, decide whether the blank should be filled in with =, <, or >, and give a clear explanation. In (a) and (b), order doesn't matter.

(a) (number of ways to choose 5 people out of 10) (number of ways to choose 6 people out of 10)

(b) (number of ways to break 10 people into 2 teams of 5) (number of ways to break 10 people into a team of 6 and a team of 4)

(c) (probability that all 3 people in a group of 3 were born on January 1) (probability that in a group of 3 people, 1 was born on each of January 1, 2, and 3)

Martin and Gale play an exciting game of "toss the coin", where they toss a fair coin until the pattern HH occurs (two consecutive Heads) or the pattern TH occurs (Tails followed immediately by Heads). Martin wins the game if and only if HH occurs before TH occurs.

(d) (probability that Martin wins) $1/2$

57. Take a deep breath before attempting this problem. In the book *Innumeracy* [20], John Allen Paulos writes:

Now for better news of a kind of immortal persistence. First, take a deep breath. Assume Shakespeare's account is accurate and Julius Caesar gasped ["Et tu, Brute!"] before breathing his last. What are the chances you just inhaled a molecule which Caesar exhaled in his dying breath?

Assume that one breath of air contains 10^{22} molecules, and that there are 10^{44} molecules in the atmosphere. (These are slightly simpler numbers than the estimates that Paulos gives; for the purposes of this problem, assume that these are exact. Of course, in reality there are many complications such as different types of molecules in the atmosphere, chemical reactions, variation in lung capacities, etc.)

Suppose that the molecules in the atmosphere now are the same as those in the atmosphere when Caesar was alive, and that in the 2000 years or so since Caesar, these molecules have been scattered completely randomly through the atmosphere. Also assume that Caesar's last breath was sampled *without* replacement but that your breathing is sampled *with* replacement (without replacement makes more sense but with replacement is easier to work with, and is a good approximation since the number of molecules in the atmosphere is so much larger than the number of molecules in one

42 breath).

Find the probability that at least one molecule in the breath you just took was shared with Caesar's last breath, and give a simple approximation in terms of e . Hint: As discussed in the math appendix, $(1 + \frac{x}{n})^n \approx e^x$ for n large.

58. A widget inspector inspects 12 widgets and finds that exactly 3 are defective. Unfortunately, the widgets then get all mixed up and the inspector has to find the 3 defective widgets again by testing widgets one by one.

(a) Find the probability that the inspector will now have to test at least 9 widgets.

(b) Find the probability that the inspector will now have to test at least 10 widgets.

59. There are 15 chocolate bars and 10 children. In how many ways can the chocolate bars be distributed to the children, in each of the following scenarios?

(a) The chocolate bars are fungible (interchangeable).

(b) The chocolate bars are fungible, and each child must receive at least one. Hint:

First give each child a chocolate bar, and then decide what to do with the rest.

(c) The chocolate bars are not fungible (it matters which particular bar goes where).

(d) The chocolate bars are not fungible, and each child must receive at least one.

Hint: The strategy suggested in (b) does not apply. Instead, consider *randomly* giving the chocolate bars to the children, and apply inclusion-exclusion.

60. Given $n \geq 2$ numbers $(a_1, a_2, \dots, a_n)$ with no repetitions, a *bootstrap sample* is a sequence $(x_1, x_2, \dots, x_n)$ formed from the a_j 's by sampling with replacement with equal probabilities. Bootstrap samples arise in a widely used statistical method known as the *bootstrap*. For example, if $n = 2$ and $(a_1, a_2) = (3, 1)$, then the possible bootstrap samples are $(3, 3), (3, 1), (1, 3)$, and $(1, 1)$.

(a) How many possible bootstrap samples are there for $(a_1, \dots, a_n)$?

(b) How many possible bootstrap samples are there for $(a_1, \dots, a_n)$, if order does not matter (in the sense that it only matters how many times each a_j was chosen, not the order in which they were chosen)?

(c) One random bootstrap sample is chosen (by sampling from $a_1, \dots, a_n$ with replacement, as described above). Show that not all unordered bootstrap samples (in the sense of (b)) are equally likely. Find an unordered bootstrap sample b_1 that is as likely as possible, and an unordered bootstrap sample b_2 that is as unlikely as possible. Let p_1 be the probability of getting b_1 and p_2 be the probability of getting b_2 (so p_i is the probability of getting the *specific* unordered bootstrap sample b_i). What is p_1/p_2 ? What is the ratio of the probability of getting an unordered bootstrap sample whose probability is p_1 to the probability of getting an unordered sample whose probability is p_2 ?

61. s There are 100 passengers lined up to board an airplane with 100 seats (with each seat assigned to one of the passengers). The first passenger in line crazily decides to sit in a randomly chosen seat (with all seats equally likely). Each subsequent passenger takes their assigned seat if available, and otherwise sits in a random available seat. What is the probability that the last passenger in line gets to sit in their assigned seat? (This is a common interview problem, and a beautiful example of the power of symmetry.)

Hint: Call the seat assigned to the j th passenger in line "seat j " (regardless of whether the airline calls it seat 23A or whatever). What are the possibilities for which seats are available to the last passenger in line, and what is the probability of each of these possibilities?

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62. In the birthday problem, we assumed that all 365 days of the year are equally likely (and excluded February 29). In reality, some days are slightly more likely as birthdays than others. For example, scientists have long struggled to understand why more babies are born 9 months after a holiday. Let $p = (p_1, p_2, \dots, p_{365})$ be the vector of birthday

probabilities, with p_j the probability of being born on the j th day of the year (February 29 is still excluded, with no offense intended to Leap Dayers).

The k th *elementary symmetric polynomial* in the variables $x_1, \dots, x_n$ is defined by

$$e_k(x_1, \dots, x_n) = \sum_{1 \leq j_1 < j_2 < \dots < j_k \leq n} x_{j_1} \dots x_{j_k}.$$

This just says to add up all of the $\binom{n}{k}$ terms we can get by choosing and multiplying k of the variables. For example, $e_1(x_1, x_2, x_3) = x_1 + x_2 + x_3$, $e_2(x_1, x_2, x_3) = x_1x_2 + x_1x_3 + x_2x_3$, and $e_3(x_1, x_2, x_3) = x_1x_2x_3$.

Now let $k \geq 2$ be the number of people.

(a) Find a simple expression for the probability that there is at least one birthday match, in terms of p and an elementary symmetric polynomial.

(b) Explain intuitively why it makes sense that $P(\text{at least one birthday match})$ is minimized when $p_j = \frac{1}{365}$ for all j , by considering simple and extreme cases.

(c) The famous *arithmetic mean-geometric mean inequality* says that for $x, y \geq 0$,

$$\frac{x+y}{2} \geq \sqrt{xy}.$$

This inequality follows from adding $4xy$ to both sides of $x^2 - 2xy + y^2 = (x - y)^2 \geq 0$.

Define $r = (r_1, \dots, r_{365})$ by $r_1 = r_2 = (p_1 + p_2)/2$, $r_j = p_j$ for $3 \leq j \leq 365$. Using the arithmetic mean-geometric mean bound and the fact, which you should verify, that

$e_k(x_1, \dots, x_n) = x_1x_2e_{k-2}(x_3, \dots, x_n) + (x_1 + x_2)e_{k-1}(x_3, \dots, x_n) + e_k(x_3, \dots, x_n)$, show that

$$P(\text{at least one birthday match} | p) \geq P(\text{at least one birthday match} | r),$$

with strict inequality if $p \neq r$, where the “given r ” notation means that the birthday probabilities are given by r . Using this, show that the value of p that minimizes the probability of at least one birthday match is given by $p_j = \frac{1}{365}$ for all j .

2

Conditional probability

We have introduced probability as a language for expressing our degrees of belief or uncertainties about events. Whenever we observe new evidence (i.e., obtain *data*), we acquire information that may affect our uncertainties. A new observation that is consistent with an existing belief could make us more sure of that belief, while a surprising observation could throw that belief into question. *Conditional probability* is the concept that addresses this fundamental question: how should we update our beliefs in light of the evidence we observe?

2.1 The importance of thinking conditionally

Conditional probability is essential for scientific, medical, and legal reasoning, since it shows how to incorporate evidence into our understanding of the world in a logical, coherent manner. In fact, a useful perspective is that *all probabilities are conditional*; whether or not it's written explicitly, there is always background knowledge (or assumptions) built into every probability.

Suppose, for example, that one morning we are interested in the event R that it will rain that day. Let $P(R)$ be our assessment of the probability of rain before looking outside. If we then look outside and see ominous clouds in the sky, then presumably our probability of rain should increase; we denote this new probability by $P(R|C)$ (read as “probability of R given C ”), where C is the event of there being ominous clouds. When we go from $P(R)$ to $P(R|C)$, we say that we are “conditioning on C ”. As the day progresses, we may obtain more and more information about the weather conditions, and we can continually update our probabilities. If we observe that events $B_1, \dots, B_n$ occurred, then we write our new conditional probability of rain given this evidence as $P(R|B_1, \dots, B_n)$. If eventually we observe that it does start raining, our conditional probability becomes 1.

Furthermore, we will see that conditioning is a very powerful problem-solving strategy, often making it possible to solve a complicated problem by decomposing it into manageable pieces with case-by-case reasoning. Just as in computer science a common strategy is to break a large problem up into bite-size pieces (or even byte-size pieces), in probability a common strategy is to reduce a complicated probability problem to a bunch of simpler conditional probability problems. In particular, we

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will discuss a strategy known as *first-step analysis*, which often allows us to obtain recursive solutions to problems where the experiment has multiple stages.

Due to the central importance of conditioning, both as the means by which we update beliefs to reflect evidence and as a problem-solving strategy, we say that

Conditioning is the soul of statistics.

2.2 Definition and intuition

Definition 2.2.1 (Conditional probability). If A and B are events with $P(B) > 0$, then the *conditional probability* of A given B , denoted by $P(A|B)$, is defined as

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B)$$

Here A is the event whose uncertainty we want to update, and B is the evidence we observe (or want to treat as given). We call $P(A)$ the *prior* probability of A and $P(A|B)$ the *posterior* probability of A (“prior” means before updating based on the evidence, and “posterior” means after updating based on the evidence).

It is important to interpret the event appearing after the vertical conditioning bar as the evidence that we have observed or that is being conditioned on: $P(A|B)$ is the probability of A given the evidence B , *not* the probability of some entity called $A|B$. As discussed in 2.4.1, there is no such event as $A|B$.

For any event A , $P(A|A) = P(A \cap A)/P(A) = 1$. Upon observing that A has occurred, our updated probability for A is 1. If this weren’t the case, we would demand a new definition of conditional probability!

Example 2.2.2 (Two cards). A standard deck of cards is shuffled well. Two cards are drawn randomly, one at a time without replacement. Let A be the event that the first card is a heart, and B be the event that the second card is red. Find $P(A|B)$ and $P(B|A)$.

Solution:

By the naive definition of probability and the multiplication rule,

$$52 \cdot 51 = 2652$$

$$P(A \cap B) = \frac{13 \cdot 25}{52 \cdot 51} = \frac{1}{102}$$

since a favorable outcome is determined by choosing any of the 13 hearts and then any of the remaining 25 red cards. Also, $P(A) = 1/4$ since the 4 suits are equally likely, and

$$P(B) = \frac{26}{52} = \frac{1}{2}$$

$$52 \cdot 51 = 2652$$

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since there are 26 favorable possibilities for the *second* card, and for each of those, the first card can be any other card (recall from Chapter 1 that chronological order is not needed in the multiplication rule).

A neater way to see that $P(B) = 1/2$ is by *symmetry*: from a vantage point before having done the experiment, the second card is equally likely to be any card in the deck.

We now have all the pieces needed to apply the definition of conditional

$$\text{probability: } P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B) = \frac{25}{102}$$

$$1/2 = \frac{51}{102}$$

$$P(B|A) = P(B \cap A)$$

$$P(A) = \frac{25}{104} = \frac{25}{51} \cdot \frac{1}{4}$$

This is a simple example, but already there are several things worth noting.

1. It's extremely important to be careful about which events to put on which side of the conditioning bar. In particular, $P(A|B) \neq P(B|A)$. The next section explores how $P(A|B)$ and $P(B|A)$ are related in general. Confusing these two quantities is called the *prosecutor's fallacy* and is discussed in Section 2.8. If instead we had defined B to be the event that the second card is a heart, then the two conditional probabilities would have been equal.

2. Both $P(A|B)$ and $P(B|A)$ make sense (intuitively and mathematically); the chronological order in which cards were chosen does not dictate which conditional probabilities we can look at. When we calculate conditional probabilities, we are considering what *information* observing one event provides about another event, not whether one event *causes* another. For further intuition, imagine that someone spreads out the cards and draws one card with their left hand and another card with their right hand, at the same time. Defining A and B based on the left hand's card and the right hand's card rather than the first card and second card would not change the structure of the problem in any important way.

3. We can also see that $P(B|A) = 25/51$ by a direct interpretation of what conditional probability means: if the first card drawn is a heart, then the remaining cards consist of 25 red cards and 26 black cards (all of which are equally likely to be drawn next), so the conditional probability of getting a red card is $25/(25 + 26) = 25/51$. It is harder to find $P(A|B)$ in this way: if we learn that the second card is red, we might think "that's nice to know, but what we really want to know is whether it's a heart!" The conditional probability results from later sections in this chapter give us methods for getting around this issue.

To shed more light on what conditional probability means, here are two intuitive interpretations.

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Intuition 2.2.3 (Pebble World). Consider a finite sample space, with the outcomes visualized as pebbles with total mass 1. Since A is an event, it is a set of pebbles, and likewise for B . Figure 2.1(a) shows an example.

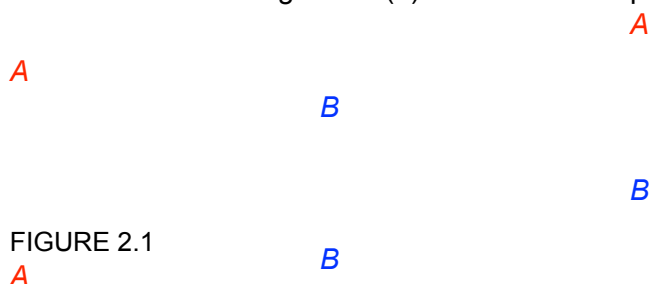


FIGURE 2.1

Pebble World intuition for $P(A|B)$. From left to right: (a) Events A and B are subsets of the sample space. (b) Because we know B occurred, get rid of the

outcomes in B^c . (c) In the restricted sample space, renormalize so the total mass is still 1.

Now suppose that we learn that B occurred. In Figure 2.1(b), upon obtaining this information, we get rid of all the pebbles in B^c because they are incompatible with the knowledge that B has occurred. Then $P(A \cap B)$ is the total mass of the pebbles remaining in A . Finally, in Figure 2.1(c), we *renormalize*, that is, divide all the masses by a constant so that the new total mass of the remaining pebbles is 1. This is achieved by dividing by $P(B)$, the total mass of the pebbles in B . The updated mass of the outcomes corresponding to event A is the conditional probability $P(A|B) = P(A \cap B)/P(B)$.

In this way, our probabilities have been updated in accordance with the observed evidence. Outcomes that contradict the evidence are discarded, and their mass is redistributed among the remaining outcomes, preserving the relative masses of the remaining outcomes. For example, if pebble 2 weighs twice as much as pebble 1 initially, and both are contained in B , then after conditioning on B it is still true that pebble 2 weighs twice as much as pebble 1. But if pebble 2 is not contained in B , then after conditioning on B its mass is updated to 0.

Intuition 2.2.4 (Frequentist interpretation). Recall that the frequentist interpretation of probability is based on relative frequency over a large number of repeated trials. Imagine repeating our experiment many times, generating a long list of observed outcomes. The conditional probability of A given B can then be thought of in a natural way: it is the fraction of times that A occurs, restricting attention to the trials where B occurs.

In Figure 2.2, our experiment has outcomes which can be written as a string of 0's and 1's; B is the event that the first digit is 1 and A is the event that the second digit is 1. Conditioning on B , we circle all the repetitions where B occurred, and then we look at the fraction of circled repetitions in which event A also occurred.

In symbols, let n_A , n_B , n_{AB} be the number of occurrences of A , B , $A \cap B$ respectively in a large number n of repetitions of the experiment. The frequentist interpretation

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is that

$$\begin{aligned} P(A) &\approx \frac{n_A}{n} \\ n, P(B) &\approx \frac{n_B}{n} \\ n, P(A \cap B) &\approx \frac{n_{AB}}{n} \end{aligned}$$

Then $P(A|B)$ is interpreted as n_{AB}/n_B , which equals $(n_{AB}/n)/(n_B/n)$. This interpretation again translates to $P(A|B) = P(A \cap B)/P(B)$.

10101010101001000010010

01010101001010000111110

11110010101010000000111

110011011010101010110

01100001110101011110011

10100110111001001001110

·
·
·

FIGURE 2.2

Frequentist intuition for $P(A|B)$. The repetitions where B occurred are circled; among these, the repetitions where A occurred are highlighted in bold. $P(A|B)$ is the long-run relative frequency of the repetitions where A occurs, within the subset of repetitions where B occurs.

For practice with applying the definition of conditional probability, let's do some more examples. The next three examples all start with the same basic scenario of a family with two children, but subtleties arise depending on the exact assumptions and the exact information we condition on.

Example 2.2.5 (Two children). Martin Gardner posed the following puzzle in the 1950s, in his column in Scientific American.

Mr. Jones has two children. The older child is a girl. What is the probability that both children are girls?

Mr. Smith has two children. At least one of them is a boy. What is the probability that both children are boys?

At first glance this problem seems like it should be a simple application of conditional probability, but for decades there have been controversies about whether or why the two parts of the problem should have different answers, and the extent to which the problem is ambiguous. Gardner gave the answers $1/2$ and $1/3$ to the two parts, respectively, which may seem paradoxical: why should it matter whether we learn the older child's gender, as opposed to just learning one child's gender?

It is important to clarify the assumptions of the problem. Several implicit assumptions are being made to obtain the answers that Gardner gave.

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- It assumes that gender is binary, so that each child can be definitively categorized as a boy or a girl. In fact, many people don't neatly fit into either of the categories "male" or "female", and identify themselves as having a non-binary gender.
- It assumes that $P(\text{boy}) = P(\text{girl})$, both for the elder child and for the younger child. In fact, in most countries slightly more boys are born than girls. For example, in the United States it is commonly estimated that 105 boys are born for every 100 girls who are born.
- It assumes that the genders of the two children are independent, i.e., knowing the elder child's gender gives no information about the younger child's gender, and vice versa. This would be unrealistic if, e.g., the children were identical twins.

Under these (admittedly problematic) simplifying assumptions, we can solve the problem as follows.

Solution:

With the assumptions listed above, the definition of conditional probability gives

$$P(\text{elder is a girl}) = 1/4$$

$$P(\text{both girls} | \text{elder is a girl}) = P(\text{both one girl})$$

$$\text{girls, elder is a girl}) P(\text{both girls} | \text{at } 1/2 = 1/2,$$

$$\text{least one girl}) = P(\text{both girls, at least$$

$$P(\text{at least one girl}) = 1/4$$

$$3/4 = 1/3.$$

(We solved the second part of the problem in terms of girls rather than boys to make it a bit easier to compare the two parts of the problem.) It may seem counterintuitive that the two results are different, since there is no reason for us to care whether the elder child is a girl as opposed to the younger child. Indeed, by symmetry,

$$P(\text{both girls} | \text{younger is a girl}) = P(\text{both girls} | \text{elder is a girl}) = 1/2.$$

However, there is no such symmetry between the conditional probabilities $P(\text{both girls} | \text{elder is a girl})$ and $P(\text{both girls} | \text{at least one girl})$. Saying that the elder child is a girl designates a *specific* child, and then the other child (the younger child) has a 50% chance of being a girl. “At least one” does *not* refer to a specific child. Conditioning on a specific child being a girl knocks away 2 of the 4 “pebbles” in the sample space $\{GG, GB, BG, BB\}$, where, for example, GB means the elder child is a girl and the younger child is a boy. In contrast, conditioning on at least one child being a girl knocks away only BB .

Example 2.2.6 (Random child is a girl). A family has two children. You randomly run into one of the two, and learn that she is a girl. With assumptions as in the previous example, what is the conditional probability that both are girls? Also assume that you are equally likely to run into either child, and that which one you run into has nothing to do with gender.

Conditional probability 51 Solution:

Intuitively, the answer should be $1/2$: imagine that the child we encountered is in front of us and the other is at home. Both being girls just says that the child who is at home is a girl, which seems to have nothing to do with the fact that the child in front of us is a girl. But let us check this more carefully, using the definition of conditional probability. This is also good practice with writing events in set notation.

Let G_1 , G_2 , and G_3 be the events that the elder, younger, and random child is a girl, respectively. By assumption, $P(G_1) = P(G_2) = P(G_3) = 1/2$. By the naive definition, or by independence as explained in Section 2.5, $P(G_1 \cap G_2) = 1/4$. Thus,

$$P(G_1 \cap G_2 | G_3) = P(G_1 \cap G_2 \cap G_3) / P(G_3) = (1/4) / (1/2) = 1/2,$$

since $G_1 \cap G_2 \cap G_3 = G_1 \cap G_2$ (if both children are girls, it guarantees that the random child is a girl).

Keep in mind though that in order to arrive at $1/2$, an assumption was needed about how the random child was selected. In statistical language, we say that we collected a *random sample*; here the sample consists of one of the two children. One of the most important principles in statistics is that it is essential to think carefully about how the sample was collected, not just stare at the raw data without understanding where they came from. To take a simple extreme case, suppose that a repressive law forbids a boy from leaving the house if he has a sister. Then “the random child is a girl” is equivalent to “at least one of the children is a girl”, so the problem reduces to the first part of Example 2.2.5.

Example 2.2.7 (A girl born in winter). A family has two children. Find the probability that both children are girls, given that at least one of the two is a girl who was born in winter. In addition to the assumptions from Example 2.2.5, assume that the four seasons are equally likely and that gender is independent of season. (This means that knowing the gender gives no information about the probabilities of the seasons, and vice versa; see Section 2.5 for much more about independence.)

Solution:

By definition of conditional probability,

$$P(\text{both girls} | \text{at least one winter girl}) = \frac{P(\text{both girls, at least one winter girl})}{P(\text{at least one winter girl})}.$$

Since the probability that a specific child is a winter-born girl is $1/8$, the denominator equals

$$P(\text{at least one winter girl}) = 1 - (7/8)^2.$$

To compute the numerator, use the fact that “both girls, at least one winter girl” is the same event as “both girls, at least one winter child”; then use the assumption

that gender and season are independent:

$$\begin{aligned} P(\text{both girls, at least one winter girl}) &= P(\text{both girls, at least one winter child}) = \\ &= (1/4)P(\text{at least one winter child}) \\ &= (1/4)(1 - P(\text{both are non-winter})) \\ &= (1/4)(1 - (3/4)^2). \end{aligned}$$

Thus,

$$\begin{aligned} P(\text{both girls} | \text{at least one winter girl}) &= \frac{(1/4)(1 - (3/4)^2)}{1 - (7/8)^2} \\ &= \frac{15/64}{7/16} = 7/15. \end{aligned}$$

At first this result seems absurd! In Example 2.2.5, the result was that the condi

tional probability of both children being girls, given that at least one is a girl, is $1/3$; why should it be any different when we learn that at least one is a winter-born girl? The point is that information about the birth season brings “at least one is a girl” closer to “a specific one is a girl”. Conditioning on more and more specific information brings the probability closer and closer to $1/2$.

For example, conditioning on “at least one is a girl who was born on a March 31 at 8:20 pm” comes very close to specifying a child, and learning information about a specific child does not give us information about the other child. The seemingly irrelevant information such as season of birth interpolates between the two parts of Example 2.2.5. Exercise 29 generalizes this example to an arbitrary characteristic that is independent of gender.

2.3 Bayes’ rule and the law of total probability

The definition of conditional probability is simple—just a ratio of two probabilities— but it has far-reaching consequences. The first consequence is obtained easily by moving the denominator in the definition to the other side of the equation.

Theorem 2.3.1 (Probability of the intersection of two events). For any events A and B with positive probabilities,

$$P(A \cap B) = P(B)P(A|B) = P(A)P(B|A).$$

This follows from taking the definition of $P(A|B)$ and multiplying both sides by $P(B)$, and then taking the definition of $P(B|A)$ and multiplying both sides by $P(A)$. At first sight this theorem may not seem very useful: it *is* the definition of conditional probability, just written slightly differently, and anyway it seems circular to use $P(A|B)$ to help find $P(A \cap B)$ when $P(A|B)$ was defined in terms of $P(A \cap B)$. But we will see that the theorem is in fact very useful, since it often turns out to be

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possible to find conditional probabilities without going back to the definition, and in such cases Theorem 2.3.1 can help us more easily find $P(A \cap B)$.

Applying Theorem 2.3.1 repeatedly, we can generalize to the intersection of n events.

Theorem 2.3.2 (Probability of the intersection of n events). For any events $A_1, \dots, A_n$ with $P(A_1, A_2, \dots, A_{n-1}) > 0$,

$$P(A_1, A_2, \dots, A_n) = P(A_1)P(A_2|A_1)P(A_3|A_1, A_2) \cdots P(A_n|A_1, \dots, A_{n-1}),$$

The commas denote intersections, e.g., $P(A_3|A_1, A_2)$ is $P(A_3|A_1 \cap A_2)$.

In fact, this is $n!$ theorems in one, since we can permute $A_1, \dots, A_n$ however we want without affecting the left-hand side. Often the right-hand side will be much easier to compute for some orderings than for others. For example,

$$P(A_1, A_2, A_3) = P(A_1)P(A_2|A_1)P(A_3|A_1, A_2) = P(A_2)P(A_3|A_2)P(A_1|A_2, A_3),$$

and there are 4 other expansions of this form too. It often takes practice and

thought to be able to know which ordering to use.

We are now ready to introduce the two main theorems of this chapter—Bayes' rule and the law of total probability—which will allow us to compute conditional probabilities in a wide range of problems. Bayes' rule is an extremely famous, extremely useful result that relates $P(A|B)$ to $P(B|A)$.

Theorem 2.3.3 (Bayes' rule).

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}.$$

This follows immediately from Theorem 2.3.1, which in turn followed immediately from the definition of conditional probability. Yet Bayes' rule has important implications and applications in probability and statistics, since it is so often necessary to find conditional probabilities, and often $P(B|A)$ is much easier to find directly than $P(A|B)$ (or vice versa).

Another way to write Bayes' rule is in terms of *odds* rather than probability.

Definition 2.3.4 (Odds). The *odds* of an event A are

$$\text{odds}(A) = P(A)/P(A^c).$$

For example, if $P(A) = 2/3$, we say the odds in favor of A are 2 to 1. (This is sometimes written as 2 : 1, and is sometimes stated as 1 to 2 odds against A ; care is needed since some sources do not explicitly state whether they are referring to odds in favor or odds against an event.) Of course we can also convert from odds back to probability:

$$P(A) = \text{odds}(A)/(1 + \text{odds}(A)).$$

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By taking the Bayes' rule expression for $P(A|B)$ and dividing it by the Bayes' rule expression for $P(A^c|B)$, we arrive at the odds form of Bayes' rule.

Theorem 2.3.5 (Odds form of Bayes' rule). For any events A and B with positive probabilities, the odds of A after conditioning on B are

$$\frac{P(A|B)}{P(A^c|B)} = \frac{P(B|A)}{P(B|A^c)} \cdot \frac{P(A)}{P(A^c)}.$$

In words, this says that the *posterior odds* $P(A|B)/P(A^c|B)$ are equal to the *prior odds* $P(A)/P(A^c)$ times the factor $P(B|A)/P(B|A^c)$, which is known in statistics as the *likelihood ratio*. Sometimes it is convenient to work with this form of Bayes' rule to get the posterior odds, and then if desired we can convert from odds back to probability.

The *law of total probability* (LOTP) relates conditional probability to unconditional probability. It is essential for fulfilling the promise that conditional probability can be used to decompose complicated probability problems into simpler pieces, and it is often used in tandem with Bayes' rule.

Theorem 2.3.6 (Law of total probability). Let $A_1, \dots, A_n$ be a partition of the

sample space S (i.e., the A_i are disjoint events and their union is S), with $P(A_i) > 0$ for all i . Then

$$P(B) = \sum_{i=1}^n P(B|A_i)P(A_i).$$

Proof. Since the A_i form a partition of S , we can decompose B as

$$B = (B \cap A_1) \cup (B \cap A_2) \cup \cdots \cup (B \cap A_n).$$

This is illustrated in Figure 2.3, where we have chopped B into the smaller pieces $B \cap A_1$ through $B \cap A_n$. By the second axiom of probability, because these pieces are disjoint, we can add their probabilities to get $P(B)$:

$$P(B) = P(B \cap A_1) + P(B \cap A_2) + \cdots + P(B \cap A_n).$$

Now we can apply Theorem 2.3.1 to each of the $P(B \cap A_i)$:

$$P(B) = P(B|A_1)P(A_1) + \cdots + P(B|A_n)P(A_n).$$

The law of total probability tells us that to get the unconditional probability of B , we can divide the sample space into disjoint slices A_i , find the conditional probability of B within each of the slices, then take a weighted sum of the conditional probabilities, where the weights are the probabilities $P(A_i)$. The choice of how to divide up the sample space is crucial: a well-chosen partition will reduce a complicated problem into simpler pieces, whereas a poorly chosen partition will only exacerbate our problems, requiring us to calculate n difficult probabilities instead of just one!

The next few examples show how we can use Bayes' rule together with the law of total probability to update our beliefs based on observed evidence.

Conditional probability 55 **B**

$$\begin{array}{ccc} & B \cap A_2 & \\ B \cap A_6 & B \cap A_3 & B \cap A_4 \\ B \cap A_1 & & B \cap A_5 \end{array}$$

$A_1 A_2 A_3 A_4 A_5 A_6$ FIGURE 2.3

The A_i partition the sample space; $P(B)$ is equal to $\sum_i P(B \cap A_i)$.

Example 2.3.7 (Random coin). You have one fair coin, and one biased coin which lands Heads with probability $3/4$. You pick one of the coins at random and flip it three times. It lands Heads all three times. Given this information, what is the probability that the coin you picked is the fair one?

Solution:

Let A be the event that the chosen coin lands Heads three times and let F be the event that we picked the fair coin. We are interested in $P(F|A)$, but it is easier to find $P(A|F)$ and $P(A|F^c)$ since it helps to know which coin we have; this suggests using Bayes' rule and the law of total probability. Doing so, we have

$$\begin{aligned}
 P(F|A) &= \frac{P(A|F)P(F)}{P(A)} \\
 &= \frac{P(A|F)P(F)}{P(A|F)P(F) + P(A|F^c)P(F^c)} \\
 &= \frac{(1/2)^3 \cdot 1/2}{(1/2)^3 \cdot 1/2 + (3/4)^3 \cdot 1/2} \\
 &\approx 0.23.
 \end{aligned}$$

Before flipping the coin, we thought we were equally likely to have picked the fair coin as the biased coin: $P(F) = P(F^c) = 1/2$. Upon observing three Heads, however, it becomes more likely that we've chosen the biased coin than the fair coin, so $P(F|A)$ is only about 0.23.

2.3.8 (Prior vs. posterior). It would *not* be correct in the calculation in the above example to say after the first step, " $P(A) = 1$ because we know A happened." It is true that $P(A|A) = 1$, but $P(A)$ is the *prior* probability of A and $P(F)$ is the *prior* probability of F —both are the probabilities before we observe any data in the

experiment. These must not be confused with *posterior* probabilities conditional on the evidence A .

Example 2.3.9 (Testing for a rare disease). A patient named Fred is tested for a disease called conditionitis, a medical condition that afflicts 1% of the population. The test result is positive, i.e., the test claims that Fred has the disease. Let D be the event that Fred has the disease and T be the event that he tests positive.

Suppose that the test is "95% accurate"; there are different measures of the accuracy of a test, but in this problem it is assumed to mean that $P(T|D) = 0.95$ and $P(T^c|D^c) = 0.95$. The quantity $P(T|D)$ is known as the *sensitivity* or *true positive rate* of the test, and $P(T^c|D^c)$ is known as the *specificity* or *true negative rate*.

Find the conditional probability that Fred has conditionitis, given the evidence provided by the test result.

Solution:

Applying Bayes' rule and the law of total probability, we have

$$\begin{aligned}
 P(D|T) &= \frac{P(T|D)P(D)}{P(T)} \\
 &= \frac{P(T|D)P(D)}{P(T|D)P(D) + P(T|D^c)P(D^c)} \\
 &= \frac{0.95 \cdot 0.01}{0.95 \cdot 0.01 + 0.05 \cdot 0.99}
 \end{aligned}$$

≈ 0.16 .

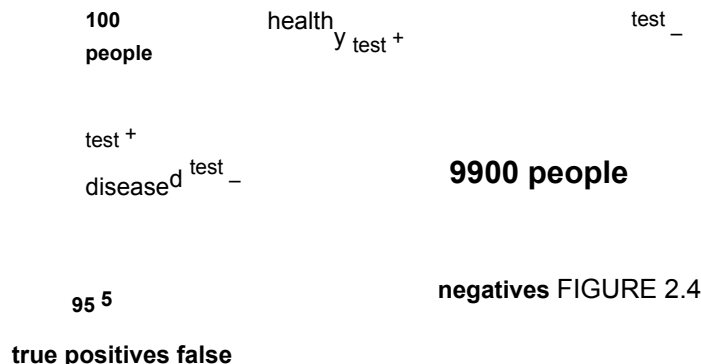
So there is only a 16% chance that Fred has conditionitis, given that he tested positive, even though the test seems to be quite reliable!

Most people find it surprising to learn that the conditional probability of having the disease given a positive test result is only 16%, even though the test is 95% accurate (see Gigerenzer and Hoffrage [13]). The key to understanding this surprisingly low posterior probability of having the disease is to realize that there are two factors at play: the evidence from the test, and our *prior information* about the prevalence of the disease.

Although the test provides evidence in favor of disease, conditionitis is also a rare condition! The conditional probability $P(D|T)$ reflects a balance between these two factors, appropriately weighing the rarity of the disease against the rarity of a mistaken test result.

For further intuition, consider a population of 10000 people as illustrated in Figure 2.4, where 100 have conditionitis and 9900 don't; this corresponds to a 1% disease rate. If we tested everybody in the population, we'd expect that out of the 100 diseased individuals, 95 would test positive and 5 would test negative. Out of the 9900 healthy individuals, we'd expect $(0.95)(9900) \approx 9405$ to test negative and 495 to test positive.

Conditional probability 57 **10000 people**



Testing for a rare disease in a population of 10000 people, where the prevalence of the disease is 1% and the true positive and true negative rates are both equal to 95%. Bubbles are not to scale.

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Now let's focus in on those individuals who test positive; that is, let's condition on a positive test result. The 95 true positives (i.e., the individuals who test positive and have the disease) are far outnumbered by the 495 false positives (i.e., the individuals who test positive despite not having the disease). So most people who test positive for the disease don't actually have the disease!

Example 2.3.10 (Six-fingered man). A crime has been committed in a certain country. The perpetrator is one (and only one) of the n men who live in the country. Initially, these n men are all deemed equally likely to be the perpetrator. An eyewitness then reports that the crime was committed by a man with six fingers on his right hand.

Let p_0 be the probability that an innocent man has six fingers on his right hand, and p_1 be the probability that the perpetrator has six fingers on his right hand, with $p_0 < p_1$. (We may have $p_1 < 1$, since eyewitnesses are not 100% reliable.) Let $a = p_0/p_1$ and $b = (1 - p_1)/(1 - p_0)$.

Rugen lives in the country. He is found to have six fingers on his right hand. (a)

Given this information, what is the probability that Rugen is the perpetrator?

(b) Now suppose that all n men who live in the country have their hands checked, and Rugen is the *only* one with six fingers on his right hand. Given this information, what is the probability that Rugen is the perpetrator?

Solution:

(a) Let R be the event that Rugen is guilty and M be the event that he has six fingers on his right hand. By Bayes' rule and LOTP,

$$P(M|R)P(R) + P(M|R^c)P(R^c) = p_1 \cdot \frac{1}{n}$$

$$P(R|M) = \frac{P(M|R)P(R)}{p_1 \cdot \frac{1}{n} + p_0 \cdot \frac{n-1}{n}} = \frac{1}{1 + a(n-1)}.$$

(b) Let N be the event that none of the men in the country other than Rugen have six fingers on their right hands. With notation as above,

$$P(R|M, N) = \frac{P(M, N|R)P(R)}{P(M, N|R)P(R) + P(M, N|R^c)P(R^c)}$$

$$\begin{aligned}
&= p_1(1 - p_0)^{n-1} \cdot \frac{1}{n} \\
& p_1(1 - p_0)^{n-1} \cdot \frac{1}{n} + p_0(1 - p_1)(1 - p_0)^{n-2} \cdot \frac{1}{n-1} \\
&= 1 \\
& 1 + ab(n - 1) \cdot
\end{aligned}$$

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2.4 Conditional probabilities are probabilities

When we condition on an event E , we update our beliefs to be consistent with this knowledge, effectively putting ourselves in a universe where we know that E occurred. Within our new universe, however, the laws of probability operate just as before. Conditional probability satisfies all the properties of probability! Therefore, any of the results we have derived about probability are still valid if we replace all unconditional probabilities with probabilities conditional on E . In particular:

- Conditional probabilities are between 0 and 1.
- $P(S|E) = 1$, $P(\emptyset|E) = 0$.

- If $A_1, A_2, \dots$ are disjoint, then $\sum_{j=1}^{\infty} P(A_j|E) = P(\bigcup_{j=1}^{\infty} A_j|E) = 1 - P(A^c|E)$.

- Inclusion-exclusion: $P(A \cup B|E) = P(A|E) + P(B|E) - P(A \cap B|E)$.

2.4.1. When we write $P(A|E)$, it does *not* mean that $A|E$ is an event and we're taking its probability; $A|E$ is not an event. Rather, $P(\cdot|E)$ is a probability function which assigns probabilities in accordance with the knowledge that E has occurred, and $P(\cdot)$ is a different probability function which assigns probabilities without regard for whether E has occurred or not. When we take an event A and plug it into the $P(\cdot)$ function, we'll get a number, $P(A)$; when we plug it into the $P(\cdot|E)$ function, we'll get another number, $P(A|E)$, which incorporates the information (if any) provided by knowing that E occurred.

To prove mathematically that conditional probabilities are probabilities, fix an event E with $P(E) > 0$, and for any event A , define $\tilde{P}(A) = P(A|E)$. This notation helps emphasize the fact that we are fixing E and treating $P(\cdot|E)$ as our new probability function. We just need to check the two axioms of probability. First,

$$\begin{aligned}
\tilde{P}(E) &= P(E|E) = P(E \cap E) = P(E) > 0 \\
\tilde{P}(\emptyset) &= P(\emptyset|E) = P(\emptyset \cap E) = 0 \\
\tilde{P}(S) &= P(S|E) = P(S \cap E) \\
\tilde{P}(E) &= 1.
\end{aligned}$$

Second, if $A_1, A_2, \dots$ are disjoint

$$\tilde{P}(A_1 \cup A_2 \cup \dots) = P((A_1 \cap E) \cup (A_2 \cap E) \cup \dots) = \sum_{j=1}^{\infty} P(A_j \cap E) = \sum_{j=1}^{\infty} P(A_j) P(E) = P(E) \sum_{j=1}^{\infty} P(A_j) = P(E).$$

So $\tilde{P}$ satisfies the axioms of probability.

Conversely, *all* probabilities can be thought of as conditional probabilities: whenever we make a probability statement, there is always some background information that we are conditioning on, even if we don't state it explicitly. Consider the rain example from the beginning of this chapter. It would be natural to base the initial probability

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of rain today, $P(R)$, on the fraction of days in the past on which it rained. But which days in the past should we look at? If it's November 1, should we only count past rainy days in autumn, thus conditioning on the season? What about conditioning on the exact month, or the exact day? We could ask the same about location: should we look at days when it rained in our exact location, or is it enough for it to have rained somewhere nearby?

In order to determine the seemingly unconditional probability $P(R)$, we actually have to make decisions about what background information to condition on! These choices require careful thought and different people may come up with different *prior* probabilities $P(R)$ (though everyone can agree on how to update based on new evidence).

Since all probabilities are conditional on background information, we can imagine that there is always a vertical conditioning bar, with background knowledge K to the right of the vertical bar. Then the unconditional probability $P(A)$ is just shorthand for $P(A|K)$; the background knowledge is absorbed into the letter P instead of being written explicitly.

To summarize our discussion in a nutshell:

Conditional probabilities are probabilities, and all probabilities are conditional.

We now state conditional forms of Bayes' rule and the law of total probability. These are obtained by taking the ordinary forms of Bayes' rule and LOTP and adding E to the right of the vertical bar everywhere.

Theorem 2.4.2 (Bayes' rule with extra conditioning). Provided that $P(A \cap E) > 0$ and $P(B \cap E) > 0$, we have

$$P(A|B, E) = \frac{P(B|A, E)P(A|E)}{P(B|E)}.$$

Theorem 2.4.3 (LOTP with extra conditioning). Let $A_1, \dots, A_n$ be a partition of S . Provided that $P(A_i \cap E) > 0$ for all i , we have

$$P(B|E) = \sum_{i=1}^n P(B|A_i, E)P(A_i|E).$$

The extra conditioning forms of Bayes' rule and LOTP can be proved similarly to

how we verified that $\tilde{P}$ satisfies the axioms of probability, but they also follow directly from the “metatheorem” that *conditional probabilities are probabilities*.

Example 2.4.4 (Random coin, continued). Continuing with the scenario from Example 2.3.7, suppose that we have now seen our chosen coin land Heads three times. If we toss the coin a fourth time, what is the probability that it will land Heads once more?

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As before, let A be the event that the chosen coin lands Heads three times, and define a new event H for the chosen coin landing Heads on the fourth toss. We are interested in $P(H|A)$. It would be very helpful to know whether we have the fair coin. LOTP with extra conditioning gives us $P(H|A)$ as a weighted average of $P(H|F, A)$ and $P(H|F^c, A)$, and within these two conditional probabilities we *do* know whether we have the fair coin:

$$\begin{aligned} P(H|A) &= P(H|F, A)P(F|A) + P(H|F^c, A)P(F^c|A) \\ &\approx \frac{1}{2} \cdot 0.23 + \frac{3}{4} \cdot (1 - 0.23) \\ &\approx 0.69. \end{aligned}$$

The posterior probabilities $P(F|A)$ and $P(F^c|A)$ are from our answer to Example 2.3.7.

An equivalent way to solve this problem is to define a new probability function $\tilde{P}$ such that for any event B , $\tilde{P}(B) = P(B|A)$. This new function assigns probabilities that are updated with the knowledge that A occurred. Then by the ordinary law of total probability,

$$\tilde{P}(H) = \tilde{P}(H|F)\tilde{P}(F) + \tilde{P}(H|F^c)\tilde{P}(F^c),$$

which is exactly the same as our use of LOTP with extra conditioning. This once again illustrates the principle that conditional probabilities are probabilities.

Example 2.4.5 (Unanimous agreement). The article “Why too much evidence can be a bad thing” by Lisa Zyga [30] says:

Under ancient Jewish law, if a suspect on trial was unanimously found guilty by all judges, then the suspect was acquitted. This reasoning sounds counterintuitive, but the legislators of the time had noticed that unanimous agreement often indicates the presence of systemic error in the judicial process.

There are n judges deciding a case. The suspect has prior probability p of being guilty. Each judge votes whether to convict or acquit the suspect. With probability s , a systemic error occurs (e.g., the defense is incompetent). If a systemic error occurs, then the judges unanimously vote to convict (i.e., all n judges vote to convict).

Whether a systemic error occurs is independent of whether the suspect is guilty. Given that a systemic error doesn’t occur and that the suspect is guilty, each judge has probability c of voting to convict, independently. Given that a systemic error doesn’t occur and that the suspect is not guilty, each judge has probability w of voting to convict, independently. Suppose that

$$0 < p < 1, 0 < s < 1, \text{ and } 0 < w < \frac{1}{2} < c < 1.$$

(a) For this part only, suppose that exactly k out of n judges vote to convict, where $k < n$. Given this information, find the probability that the suspect is guilty.

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(b) Now suppose that all n judges vote to convict. Given this information, find the probability that the suspect is guilty.

(c) Is the answer to (b), viewed as a function of n , an increasing function? Give a short, intuitive explanation in words.

Solution:

(a) Since $k < n$, a systemic error didn't occur. We will implicitly condition on this in this part. Let G be the event that the suspect is guilty and X be the number of judges who vote to convict. Using Bayes' rule, LOTP, and the Binomial PMF,

$$P(X = k) = pc^k(1 - c)^{n-k}$$

$$P(G|X = k) = \frac{P(X = k|G)P(G)}{P(X = k|G)P(G) + P(X = k|G^c)P(G^c)}$$

$$= \frac{pc^k(1 - c)^{n-k} + (1 - p)w^k(1 - w)^{n-k}}{pc^k(1 - c)^{n-k} + (1 - p)w^k(1 - w)^{n-k}}.$$

(b) Let U be the event $X = n$ and B be the event that a systemic error occurs. Then

By LOTP with extra conditioning,

$$P(U) = pP(U|G)$$

$$P(G|U) = \frac{P(U|G)P(G)}{pP(U|G) + (1 - p)P(U|G^c)}.$$

$$P(U|G) = P(U|G, B)P(B|G) + P(U|G, B^c)P(B^c|G) = s + (1 - s)c^n, P(U|G^c)$$

$$= P(U|G^c, B)P(B|G^c) + P(U|G^c, B^c)P(B^c|G^c) = s + (1 - s)w^n.$$

Thus,

$$P(G|U) = \frac{p(s + (1 - s)c^n)}{p(s + (1 - s)c^n) + (1 - p)(s + (1 - s)w^n)}.$$

(c) No, since a large value of n yields a high chance of systemic error, and if a systemic error occurs then the judges' votes are uninformative about whether the suspect is guilty. The answer to (b) reverts to the prior probability p as $n \rightarrow \infty$.

We often want to condition on more than one piece of information, and we now have several ways of doing that. For example, here are some approaches for finding $P(A|B, C)$:

1. We can think of B, C as the single event $B \cap C$ and use the definition of conditional probability to get

$$P(A|B, C) = \frac{P(A, B, C)}{P(B, C)}.$$

$$P(B, C)$$

This is a natural approach if it's easiest to think about B and C in tandem. We can then try to evaluate the numerator and denominator. For example, we can use LOTP in both the numerator and the denominator, or we can write the numerator as $P(B, C|A)P(A)$ (which would give us a version of Bayes' rule) and use LOTP to help with the denominator.

2. We can use Bayes' rule with extra conditioning on C to get

$$P(A|B, C) = P(B|A, C)P(A|C)$$

$$P(B|C) \cdot$$

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This is a natural approach if we want to think of everything in our problem as being conditioned on C .

3. We can use Bayes' rule with extra conditioning on B to get

$$P(A|B, C) = P(C|A, B)P(A|B)$$

$$P(C|B) \cdot$$

This is the same as the previous approach, except with the roles of B and C swapped. We mention it separately just to emphasize that it's a bad idea to plug into a formula without thinking about which event should play which role.

It is both challenging and powerful that there are a variety of ways to approach this kind of conditioning problem.

2.5 Independence of events

We have now seen several examples where conditioning on one event changes our beliefs about the probability of another event. The situation where events provide no information about each other is called *independence*.

Definition 2.5.1 (Independence of two events). Events A and B are *independent* if

$$P(A \cap B) = P(A)P(B).$$

If $P(A) > 0$ and $P(B) > 0$, then this is equivalent to

$$P(A|B) = P(A),$$

and also equivalent to $P(B|A) = P(B)$.

In words, two events are independent if we can obtain the probability of their intersection by multiplying their individual probabilities. Alternatively, A and B are independent if learning that B occurred gives us no information that would change our probabilities for A occurring (and vice versa).

Note that independence is a *symmetric relation*: if A is independent of B , then B is independent of A .

2.5.2. Independence is completely different from disjointness. If A and B are disjoint, then $P(A \cap B) = 0$, so disjoint events can be independent only if $P(A) = 0$ or $P(B) = 0$. Knowing that A occurs tells us that B definitely did not occur, so A clearly conveys information about B , meaning the two events are not independent (except if A or B already has zero probability).

Intuitively, it makes sense that if A provides no information about whether or not B occurred, then it also provides no information about whether or not B^c occurred.

We now prove a handy result along those lines.

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Proposition 2.5.3. If A and B are independent, then A and B^c are independent, A^c and B are independent, and A^c and B^c are independent.

Proof. Let A and B be independent. We will first show that A and B^c are independent. If $P(A) = 0$, then A is independent of every event, including B^c . So assume $P(A) > 0$. Then

$$P(B^c|A) = 1 - P(B|A) = 1 - P(B) = P(B^c),$$

so A and B^c are independent. Swapping the roles of A and B , we have that A^c and B are independent. Using the fact that A , B independent implies A , B^c independent, with A^c playing the role of A , we also have that A^c and B^c are independent.

We also often need to talk about independence of three or more events.

Definition 2.5.4 (Independence of three events). Events A , B , and C are said to be *independent* if all of the following equations hold:

$$\begin{aligned}P(A \cap B) &= P(A)P(B), \\P(A \cap C) &= P(A)P(C), \\P(B \cap C) &= P(B)P(C), \\P(A \cap B \cap C) &= P(A)P(B)P(C).\end{aligned}$$

If the first three conditions hold, we say that A , B , and C are *pairwise independent*. Pairwise independence does *not* imply independence: it is possible that just learning about A or just learning about B is of no use in predicting whether C occurred, but learning that *both* A and B occurred could still be highly relevant for C . Here is a simple example of this distinction.

Example 2.5.5 (Pairwise independence doesn't imply independence). Consider two fair, independent coin tosses, and let A be the event that the first is Heads, B the event that the second is Heads, and C the event that both tosses have the same result. Then A , B , and C are pairwise independent but not independent, since $P(A \cap B \cap C) = 1/4$ while $P(A)P(B)P(C) = 1/8$. The point is that just knowing about A or just knowing about B tells us nothing about C , but knowing what happened with *both* A and B gives us information about C (in fact, in this case it gives us perfect information about C).

On the other hand, $P(A \cap B \cap C) = P(A)P(B)P(C)$ does not imply pairwise independence; this can be seen quickly by looking at the extreme case $P(A) = 0$, when the equation becomes $0 = 0$, which tells us nothing about B and C .

We can define independence of any number of events similarly. Intuitively, the idea is that knowing what happened with any particular subset of the events gives us no information about what happened with the events not in that subset.

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Definition 2.5.6 (Independence of many events). For n events $A_1, A_2, \dots, A_n$ to

be *independent*, we require any pair to satisfy $P(A_i \cap A_j) = P(A_i)P(A_j)$ (for $i \neq j$), any triplet to satisfy $P(A_i \cap A_j \cap A_k) = P(A_i)P(A_j)P(A_k)$ (for i, j, k distinct), and similarly for all quadruplets, quintuplets, and so on. This can quickly become unwieldy, but later we will discuss other ways to think about independence. For infinitely many events, we say that they are independent if every finite subset of the events is independent.

Conditional independence is defined analogously to independence.

Definition 2.5.7 (Conditional independence). Events A and B are said to be *conditionally independent* given E if $P(A \cap B|E) = P(A|E)P(B|E)$.

2.5.8. It is easy to make terrible blunders stemming from confusing independence and conditional independence. Two events can be conditionally independent given E , but not independent given E^c . Two events can be conditionally independent given E , but not independent. Two events can be independent, but not conditionally independent given E .

In particular, $P(A, B) = P(A)P(B)$ does *not* imply $P(A, B|E) = P(A|E)P(B|E)$; we can't just insert "given E " everywhere, as we did in going from LOTP to LOTP with extra conditioning. This is because LOTP *always* holds (it is a consequence of the axioms of probability), whereas $P(A, B)$ may or may not equal $P(A)P(B)$, depending on what A and B are.

The next few examples illustrate these distinctions. Great care is needed in working with conditional probabilities and conditional independence!

Example 2.5.9 (Conditional independence given E vs. given E^c). Suppose there are two types of classes: good classes and bad classes. In good classes, if you work hard, you are very likely to get an A. In bad classes, the professor randomly assigns grades to students regardless of their effort. Let G be the event that a class is good, W be the event that you work hard, and A be the event that you receive an A. Then W and A are conditionally independent given G^c , but they are not conditionally independent given G .

Example 2.5.10 (Conditional independence doesn't imply independence). Returning once more to the scenario from Example 2.3.7, suppose we have chosen either a fair coin or a biased coin with probability 3/4 of Heads, but we do not know which one we have chosen. We flip the coin a number of times. Conditional on choosing the fair coin, the coin tosses are independent, with each toss having probability 1/2 of Heads. Similarly, conditional on choosing the biased coin, the tosses are independent, each with probability 3/4 of Heads.

However, the coin tosses are not unconditionally independent, because if we don't know which coin we've chosen, then observing the sequence of tosses gives us information about whether we have the fair coin or the biased coin in our hand. This in turn helps us to predict the outcomes of future tosses from the same coin.

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To state this formally, let F be the event that we've chosen the fair coin, and let A_1 and A_2 be the events that the first and second coin tosses land Heads. Conditional on F , A_1 and A_2 are independent, but A_1 and A_2 are not unconditionally independent because A_1 provides information about A_2 .

Example 2.5.11 (Independence doesn't imply conditional independence). My friends Alice and Bob are the only two people who ever call me on the phone. Each day, they decide independently whether to call me that day. Let A be the event that Alice calls me next Friday and B be the event that Bob calls me next Friday. Assume A and B are unconditionally independent with $P(A) > 0$ and $P(B) > 0$.

However, given that I receive exactly one call next Friday, A and B are no longer independent: the call is from Alice if and only if it is not from Bob. In other words, letting C be the event that I receive exactly one call next Friday, $P(B|C) > 0$ while $P(B|A, C) = 0$, so A and B are not conditionally independent given C .

Example 2.5.12. (Why is the baby crying?) A certain baby cries if and only if she is hungry, tired, or both. Let C be the event that the baby is crying, H be the event that she is hungry, and T be the event that she is tired. Let $P(C) = c$, $P(H) = h$, and $P(T) = t$, where none of c , h , t are equal to 0 or 1. Let H and T be independent.

(a) Find c , in terms of h and t .

(b) Find $P(H|C)$, $P(T|C)$, and $P(H, T|C)$.

(c) Are H and T conditionally independent given C ? Explain in two ways: algebraically using the quantities from (b), and with an intuitive explanation in words.

Solution:

(a) Since H and T are independent, we have

$$P(C) = P(H \cup T) = P(H) + P(T) - P(H \cap T) = h + t - ht.$$

(b) By Bayes' rule,

$$P(C) = \frac{h}{c}$$

$$P(H|C) = P(C|H)P(H)$$

$$P(C) = \frac{t}{c}$$

$$P(T|C) = P(C|T)P(T)$$

$$P(H, T|C) = P(C|H, T)P(H, T)$$

$$P(C) = \frac{ht}{c}$$

(c) No, H and T are not conditionally independent given C , since

$$P(H, T|C) = \frac{ht}{c} < \frac{ht}{c^2} = P(H|C)P(T|C).$$

We can also see intuitively why they are not conditionally independent given C : if the baby is crying but not hungry, she must be tired.

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An important property of Bayes' rule is that it is *coherent*: if we receive multiple pieces of information and wish to update our probabilities to incorporate all the information, it does not matter whether we update sequentially, taking each piece of evidence into account one at a time, or simultaneously, using all the evidence at once. Suppose, for example, that we're conducting a weeklong experiment that yields data at the end of each day. We could use Bayes' rule every day to update our probabilities based on the data from that day. Or we could go on vacation for the week, come back on Friday afternoon, and update using the entire week's worth of data. Either method will give the same result.

Let's look at a concrete application of this principle.

Example 2.6.1 (Testing for a rare disease, continued). Fred, who tested positive for conditionitis in Example 2.3.9, decides to get tested a second time. The new test is independent of the original test (given his disease status) and has the same sensitivity and specificity. Unfortunately for Fred, he tests positive a second time. Find the probability that Fred has the disease, given the evidence, in two ways: in one step, conditioning on both test results simultaneously, and in two steps, first updating the probabilities based on the first test result, and then updating again based on the second test result.

Solution:

Let D be the event that he has the disease, T_1 that the first test result is positive, and T_2 that the second test result is positive. In Example 2.3.9, we used Bayes' rule and the law of total probability to find $P(D|T_1)$. Another quick solution uses the odds form of Bayes' rule:

$$\frac{P(D^c|T_1)}{P(D|T_1)} = \frac{P(D^c)}{P(D)} \frac{P(T_1|D^c)}{P(T_1|D)} = \frac{1}{99} \cdot \frac{0.95}{0.05} \approx 0.19.$$

Since $P(D|T_1)/(1 - P(D|T_1)) = 0.19$, we have $P(D|T_1) = 0.19/(1 + 0.19) \approx 0.16$, in agreement with our answer from before. The odds form of Bayes' rule is faster in this case because we don't need to compute the unconditional probability $P(T_1)$ in the denominator of the ordinary Bayes' rule. Now, again using the odds form of Bayes' rule, let's find out what happens if Fred tests positive a second time.

One-step method: Updating based on both test results at once, we

$$\begin{aligned} &\text{have } P(D|T_1 \cap T_2) \\ &\frac{P(D^c|T_1 \cap T_2)}{P(D|T_1 \cap T_2)} = \frac{P(D^c)}{P(D)} \frac{P(T_1 \cap T_2|D^c)}{P(T_1 \cap T_2|D)} \\ &= \frac{1}{99} \cdot \frac{0.95^2}{0.05^2} \approx 3.646, \end{aligned}$$

which corresponds to a probability of 0.78.

Two-step method: After the first test, the posterior odds of Fred having the

disease are

$$\frac{P(D^c|T_1)}{P(D|T_1)} = \frac{1}{99} \cdot \frac{0.95}{0.05}$$

$$0.05 \approx 0.19,$$

from the above. These posterior odds become the new prior odds, and then updating based on the second test gives

$$\begin{aligned} & \frac{P(D^c|T_1 \cap T_2)}{P(D|T_1 \cap T_2)} = \frac{P(D^c|T_1)}{P(D|T_1)} \cdot \frac{P(T_2|D^c, T_1)}{P(T_2|D, T_1)} \\ & = \frac{1}{99} \cdot \frac{0.95}{0.05} \cdot \frac{0.05}{0.95} \\ & = \frac{1}{99} \cdot \frac{0.05}{0.95} \end{aligned}$$

which is the same result as above.

Note that with a second positive test result, the probability that Fred has the disease jumps from 0.16 to 0.78, making us much more confident that Fred is actually afflicted with conditionitis. The moral of the story is that getting a second opinion is a good idea!

2.7 Conditioning as a problem-solving tool

Conditioning is a powerful tool for solving problems because it lets us engage in *wishful thinking*: when we encounter a problem that would be made easier if only we knew whether E happened or not, we can condition on E and then on E^c , consider these possibilities separately, then combine them using LOTP.

2.7.1 Strategy: condition on what you wish you knew

Example 2.7.1 (Monty Hall). On the game show Let's Make a Deal, hosted by Monty Hall, a contestant chooses one of three closed doors, two of which have a goat behind them and one of which has a car. Monty, who knows where the car is, then opens one of the two remaining doors. The door he opens always has a goat behind it (he never reveals the car!). If he has a choice, then he picks a door at random with equal probabilities. Monty then offers the contestant the option of switching to the other unopened door. If the contestant's goal is to get the car, should she switch doors?

Solution:

Let's label the doors 1 through 3. Without loss of generality, we can assume the contestant picked door 1 (if she didn't pick door 1, we could simply relabel the doors, or rewrite this solution with the door numbers permuted). Monty opens a door, revealing a goat. As the contestant decides whether or not to switch to the remaining unopened door, what does she really wish she knew? Naturally, her decision would be a lot easier if she knew where the car was! This suggests that we should condition on the location of the car. Let C_i be the event that the car is behind door i , for $i = 1, 2, 3$. By the law of total probability,

$$P(\text{get car}) = P(\text{get car}|C_1) \cdot \frac{1}{3} + P(\text{get car}|C_2) \cdot \frac{1}{3} + P(\text{get car}|C_3) \cdot \frac{1}{3}.$$

Suppose the contestant employs the switching strategy. If the car is behind door 1, then switching will fail, so $P(\text{get car}|C_1) = 0$. If the car is behind door 2 or 3, then because Monty always reveals a goat, the remaining unopened door must contain the car, so switching will succeed. Thus,

$$P(\text{get car}) = 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} = \frac{2}{3},$$

so the switching strategy succeeds $2/3$ of the time. The contestant should switch to the other door.

Figure 2.5 is a tree diagram of the argument we have just outlined: using the switching strategy, the contestant will win as long as the car is behind doors 2 or 3, which has probability $2/3$. We can also give an intuitive frequentist argument in favor of switching. Imagine playing this game 1000 times. Typically, about 333 times your initial guess for the car's location will be correct, in which case switching will fail. The other 667 or so times, you will win by switching.

There's a subtlety though, which is that when the contestant chooses whether to switch, she also knows which door Monty opened. We showed that the *unconditional* probability of success is $2/3$ (when following the switching strategy), but let's also show that the *conditional* probability of success for switching, given the information that Monty provides, is also $2/3$.

Let M_j be the event that Monty opens door j , for $j = 2, 3$. Then $P(\text{get$

$$\text{car}) = P(\text{get car}|M_2)P(M_2) + P(\text{get car}|M_3)P(M_3),$$